SOFTWARE PROCESS MODELS CCSW-314

جامعة جدة
University of Jeddah

MALI Application

YOUR SMART PERSONAL FINANCE MANAGER

Supervisor:

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Sprint Zero

Project Description

Mali is a simple and smart financial mobile app that helps users manage their monthly budget, track daily expenses, and set saving goals with clear progress indicators.

Problem Definition

Many individuals, especially university students and new employees, struggle to manage their monthly expenses and often spend money randomly without proper planning. This leads to overspending and difficulty in achieving saving goals.

Proposed Solution

The proposed solution is a mobile app that enables users to:

- Create a monthly budget divided into categories
- Record income and expenses on a daily basis
- Set saving goals and monitor their progress with visual indicators

Scope of the Project

The project will focus on three main features:

- **Budget creation and category allocation**
- **Daily tracking of income and expenses**
- **Saving goals tracking with progress monitoring**

Users

- University students
- Employees
- Any individual who wants to manage personal finance

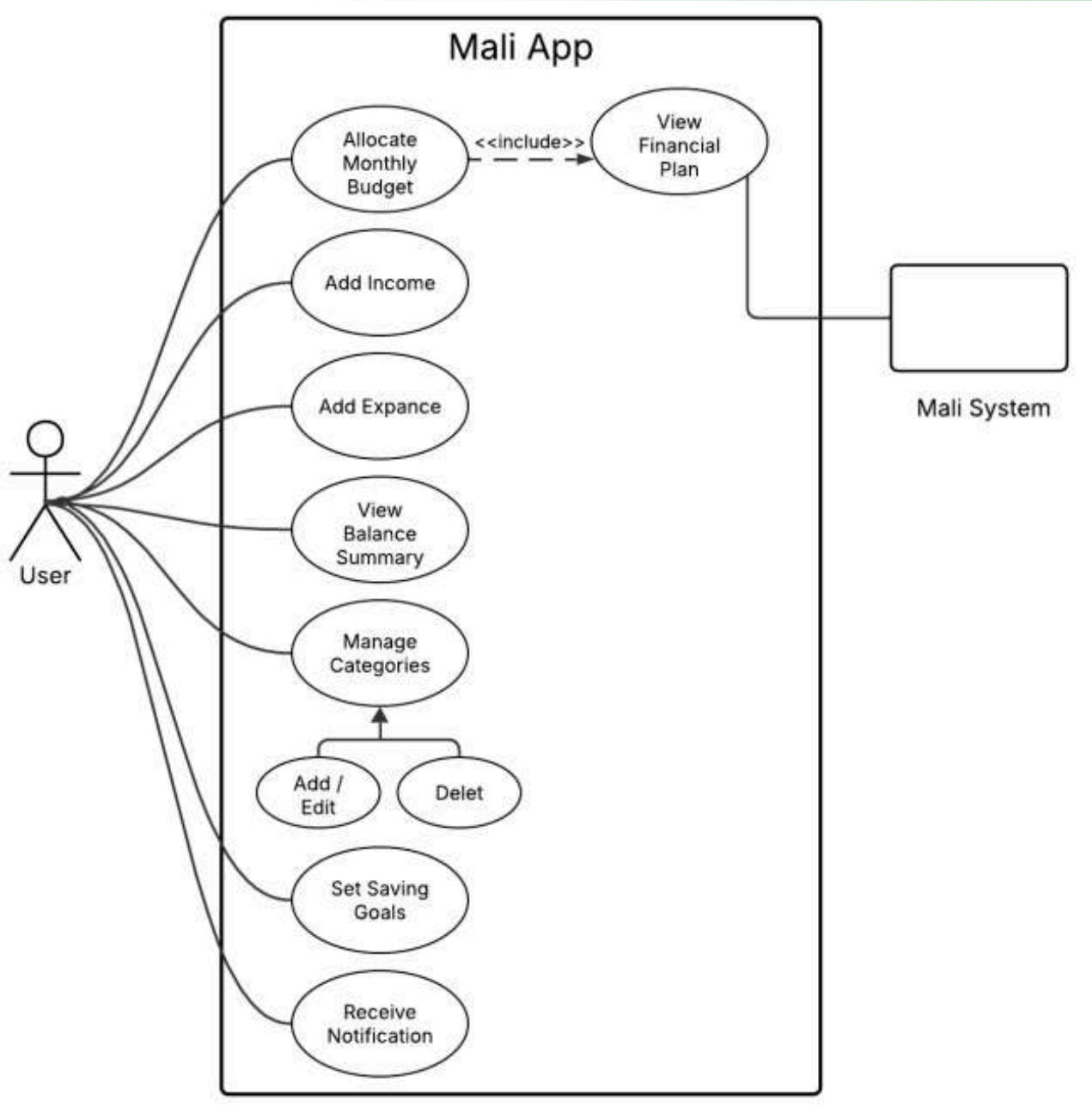
Tools

- Planning: Notion
- Communication Tool: Slack
- Version Control: GitHub
- Design & Modeling: Lucidchart
- Programming Language & IDE: Flutter (Dart) With VS Code

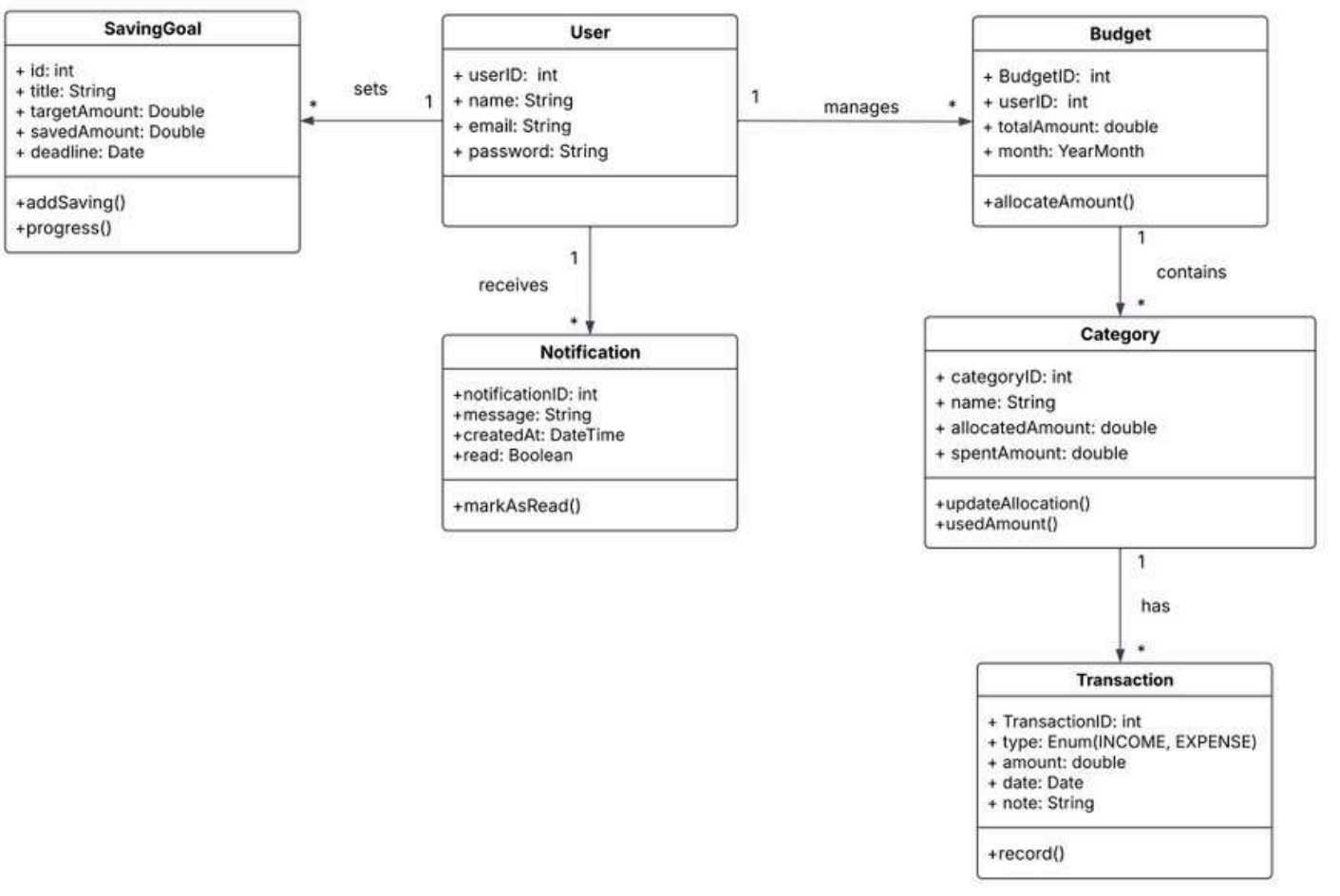
✦ Functional / Non Functional Requirements

Type	Requirement	Priority
FR-1	The user shall be able to enter their personal information and then choose the appropriate monthly budget distribution based on predefined categories.	HIGH
FR-2	The user shall be able to enter their monthly income and track daily expenses. The system shall update and display the remaining balance accordingly.	HIGH
FR-3	The user shall be able to set saving goals and track their progress towards achieving these goals.	HIGH
FR-4	The system shall provide a summary view of the monthly balance, showing the total income, total expenses, remaining balance, and progress towards savings goals.	Medium
FR-5	The user shall be able to add, edit, or remove budget categories	Medium
FR-6	The system shall send alerts or reminders when the user is close to exceeding a budget category limit or when it is time to review savings progress.	Low
NFR-1	Usability: The app must have a simple and easy-to-use interface.	HIGH
NFR-2	Performance: The app should be fast and smooth with no delays.	HIGH
NFR-3	Security: All financial data must be encrypted and protected.	HIGH
NFR-4	Availability: The app should work reliably and offer basic offline use.	Low

✦ Usecase


Mali App
Allocate Monthly Budget
<<include>>
View Financial Plan
Add Income
Add Expance
View Balance Summary
Manage Categories
Add / Edit
Delet
Set Saving Goals
Receive Notification
User
Mali System

✦ Class Diagram



Sprint 1

Project Initiation

Project Initiation

Project Name: Mali App

Project Members: Joury Almutairi -Hanan Alnejaimi -Reema Alzahrani-Layan Aljohani -Elaf Althubaiti

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Project Description:

Mali is a simple budgeting app that helps users manage their monthly income and spending. In Sprint 1, the focus was to allow users to enter their personal information and set up their monthly budget by dividing it into predefined categories such as Basics, Transport, and Savings.

Prioritized Functional and non-functional requirements:

Functional Requirement (FR-1):

The user shall be able to enter their personal information and then choose the appropriate monthly budget distribution based on predefined categories.

Non-Functional Requirements:

1. NFR-1: Usability – The app must have a clean and easy-to-use interface.
2. NFR-2: Performance – Data entry and saving should be fast and smooth.
3. NFR-3: Security – User information must be securely stored.

EXAMPLE IF YOU CHOOSE SCRUM: Story Backlog :

Component Name: Personal Information

Story Name: User Profile Setup
Story Sequence No: 001

Story Short Description:
Users can complete their profile by entering their personal information such as name, email, and monthly income.

Story Long Description:
Users shall be able to provide their personal details including name, email address, phone number, age range, and employment status.
The system shall store this information securely and use it to personalize the user experience.

Project Initiation

Once the information is submitted, the app will display a confirmation message and proceed to the next step of setting up the monthly budget.

Component Name: Budget Allocation

Story Name: Budget Distribution Setup
Story Sequence No: 002

Story Short Description:
Users can divide their monthly income into different budget categories such as Basics, Transport, and Savings.

Story Long Description:
After entering personal information, users shall be able to allocate their total monthly income across predefined categories (e.g., Basics, Transport, Savings).
The system shall calculate the percentage or amount assigned to each category and display a summary on the home screen.
This setup allows the user to visualize how their budget is distributed and supports future expense tracking.

Prioritize stories and define sprints:

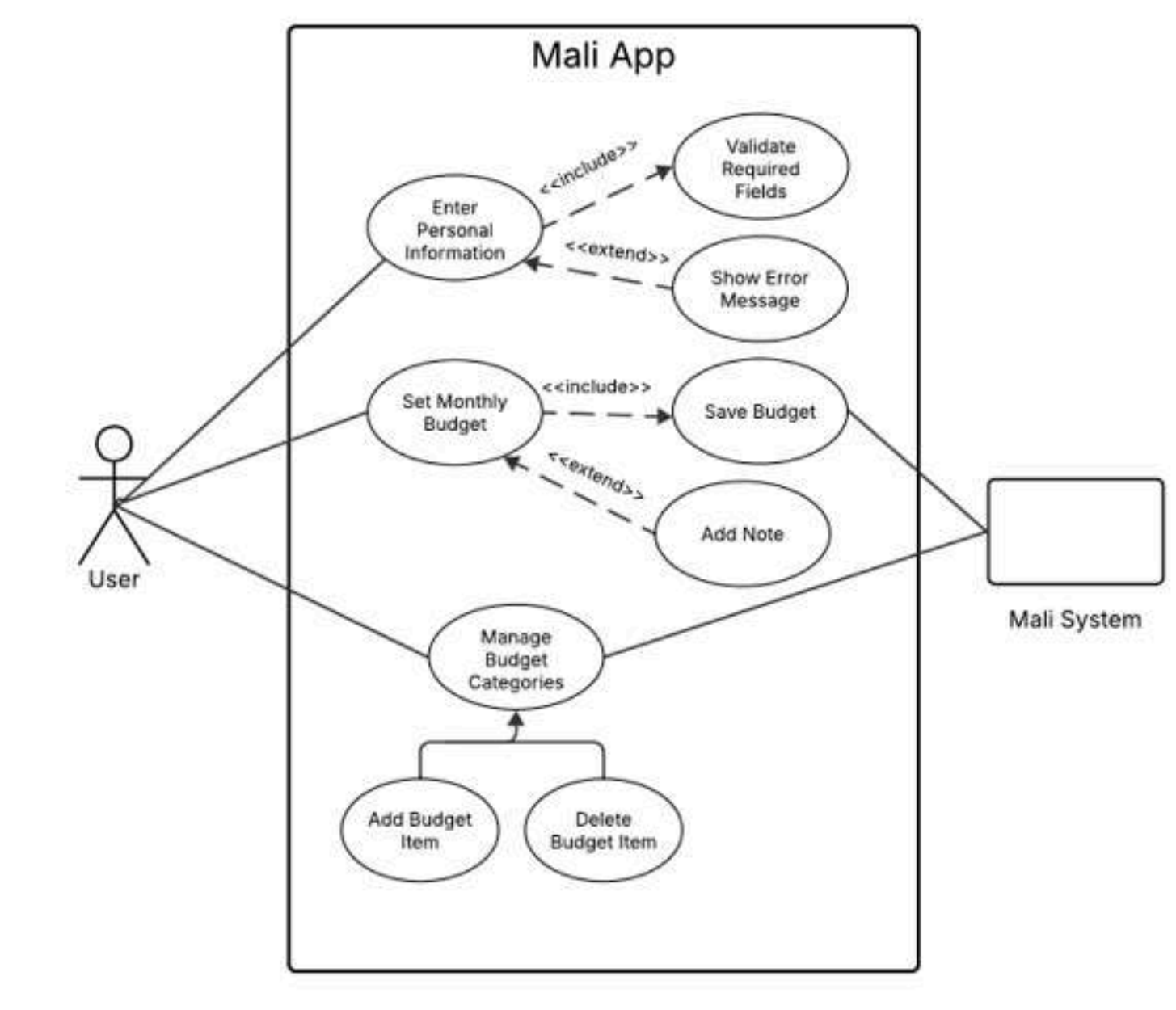
Sprint 1 Goal:

Develop the main feature that allows users to enter their information and allocate their monthly budget into categories.

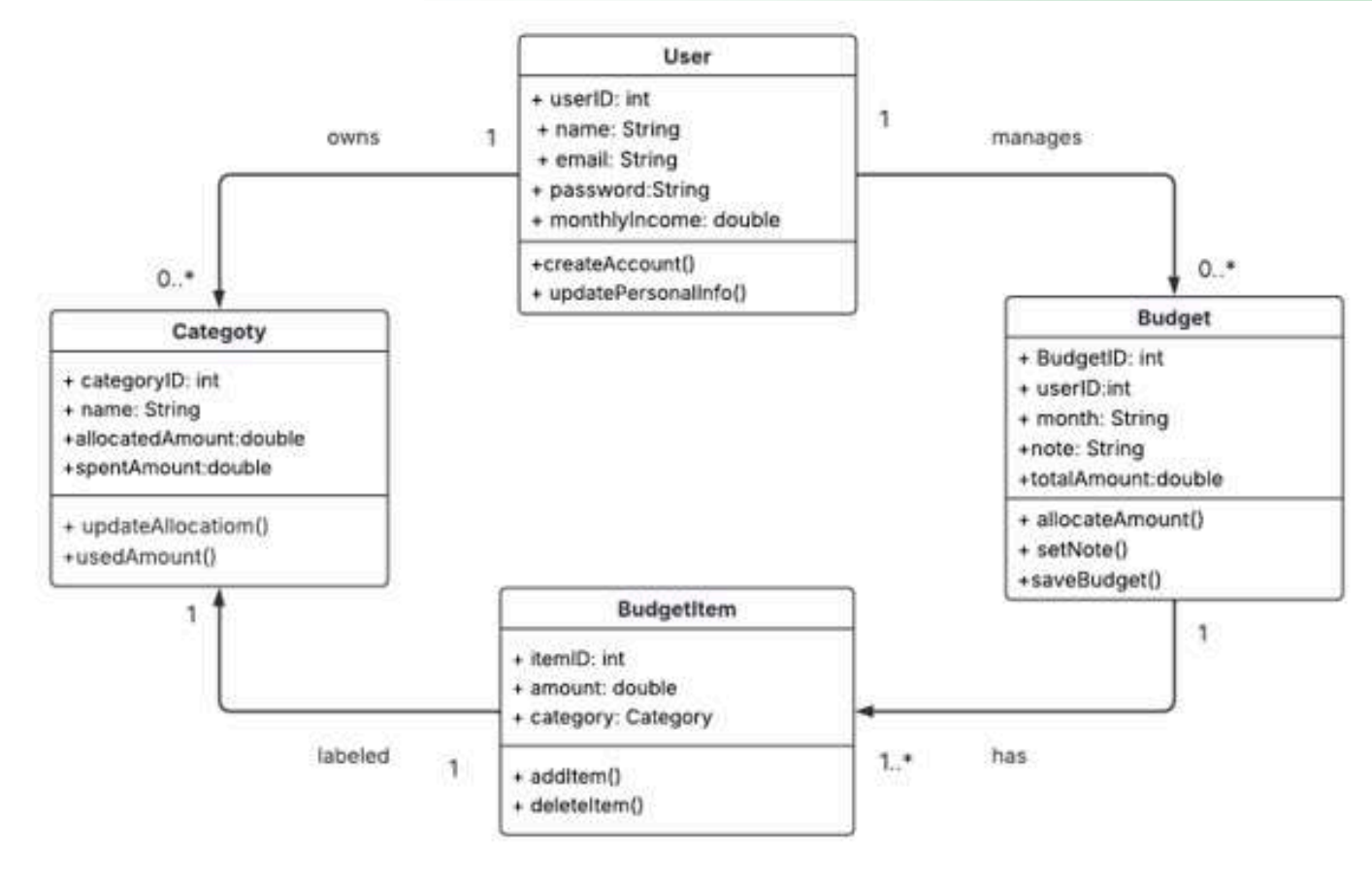
Sprint 1 Deliverables:

- Personal Information screen.
- Home screen with income summary.
- Budget allocation screen.
- Input validation and saving logic.
- Basic testing for all functions.

✦ Usecase


Mali App
Enter Personal Information
<<include>>
Validate Required Fields
<<extend>>
Show Error Message
Set Monthly Budget
<<include>>
Save Budget
<<extend>>
Add Note
Manage Budget Categories
Add Budget Item
Delete Budget Item
User
Mali System

✦ Class Diagram



Sprint Meeting(1)

Sprint Meeting(s)

Project Name: Mali

Project Members: Hanan Alnejaimi - Reema Alzahrani - Layan Aljohani - Joury Almutairi - Elaf Althubaiti

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Sprint #1 Stand up Meeting - [25/9/2025]

Sprint Duration: 3 weeks

Scrum Master: Dr. Malak Alharbi

Client: Dr. Malak Alharbi

Pair Programmers: Layan Aljohani & Reema Alzahrani

Stories:

Component Name	Story Sequence Number	Use Cases (e.g., functionalities)
Personal Information	001	Users can enter and validate their personal information.
Budget Allocation	002	Users can set and save their monthly budget, add notes, and manage budget categories by adding or deleting items to customize their budget distribution.

Follow-up meeting questions:

1. **What has been completed since the last meeting?** We designed the main user interface screens and started coding the interfaces
2. **What are you going to be working on next?** Implement the core functions and connect the firebase
3. **Do you have any issues/impediments?** Yes, we faced issues in installing flutter

Scrum's Master comments based on the above questions:

Sprint Meeting(2)

Sprint Meeting(s)

Project Name: Mali

Project Members Hanan Alnejaimi - Reema Alzahrani - Layan Aljohani - Joury Almutairi - Elaf Althubaiti

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Sprint #1 Stand up Meeting - [5/10/2025]

Sprint Duration: 3 weeks

Scrum Master: Dr. Malak Alharbi

Client: Dr. Malak Alharbi

Pair Programmers: Layan Aljohani & Reema Alzahrani

Stories:

Component Name	Story Sequence Number	Use Cases (e.g., functionalities)
Personal Information	001	Users can enter and validate their personal information.
Budget Allocation	002	Users can set and save their monthly budget, add notes, and manage budget categories by adding or deleting items to customize their budget distribution.

Follow-up meeting questions:

1. **What has been completed since the last meeting?** We solved issues related to downloading flutter and connected the budget setup to Firebase for saving user data.
2. **What are you going to be working on next?** We will work on the budget tracking page, allowing users to add and delete expense items with automatic balance updates.
3. **Do you have any issues/impediments?** Yes, we faced issues in connecting Firebase.

Scrum's Master comments based on the above questions:

Sprint Meeting(3)

Sprint Meeting(s)

Project Name: Mali

Project Members: Hanan Alnejaimi - Reema Alzahrani - Layan Aljohani - Joury Almutairi - Elaf Althubaiti

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Sprint #1 Stand up Meeting - [9/10/2025]

Sprint Duration: 3 weeks

Scrum Master: Dr. Malak Alharbi

Client: Dr. Malak Alharbi

Pair Programmers: Layan Aljohani & Reema Alzahrani

Stories:

Component Name	Story Sequence Number	Use Cases (e.g., functionalities)
Personal Information	001	Users can enter and validate their personal information.
Budget Allocation	002	Users can set and save their monthly budget, add notes, and manage budget categories by adding or deleting items to customize their budget distribution.

Follow-up meeting questions:

1. **What has been completed since the last meeting?** We completed the budget tracking functionality and tested the automatic balance updates successfully.
2. **What are you going to be working on next?** We will start preparing for Sprint 2
3. **Do you have any issues/impediments?** No

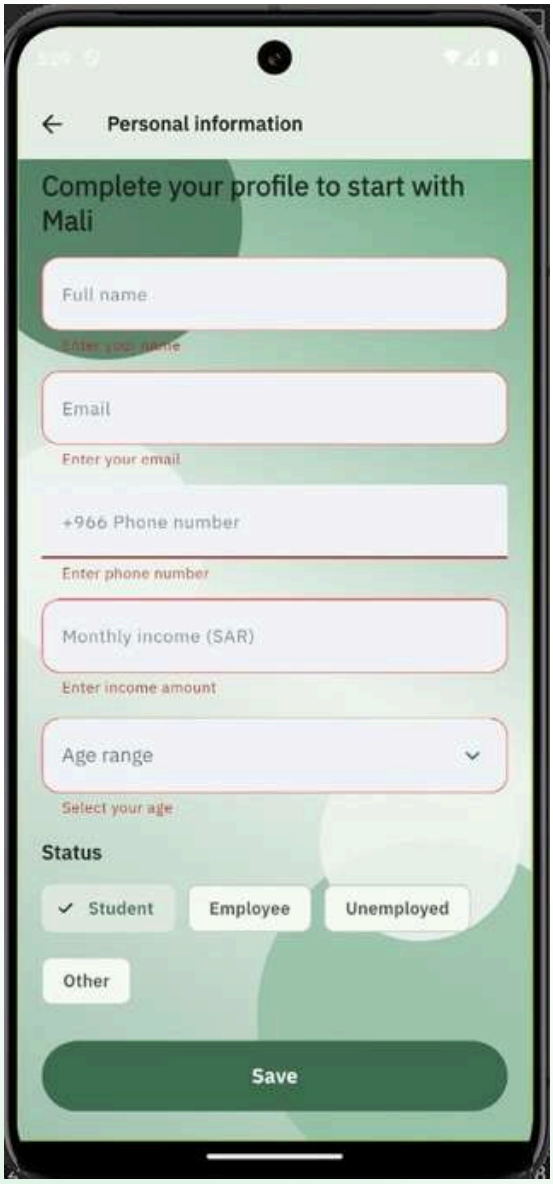
Scrum's Master comments based on the above questions:

Sprint Test Case1

Sprint 1 / Test Cases – Personal Information

Test Case No.	Test Case Description	Expected Results	Outcome (Pass / Fail / Comments)
TC01	User enters invalid email format and taps Save	Error message appears under email field: “Invalid email”.	pass
TC02	User enters non-numeric income and taps Save	Error message appears under income field: “Enter a valid number”.	pass
TC03	User leaves Age range not selected and taps Save	Error message appears under the age dropdown: “Select your age”.	pass
TC04	User leaves the Phone number field empty and taps Save	Error message appears under the phone field: “Enter your phone number”.	pass
TC05	User enters phone shorter than 9 digits (after +966) and taps Save	Error message appears: “Phone must be 9 digits”.	pass
TC06	User enters phone number that does not start with 5 and taps Save	Error message appears: “Phone number must start with 5 and be 9 digits long”.	pass
TC07	User selects a different Status chip (e.g., from Student to Employee)	The selected chip visually updates and the selected value is stored internally.	pass
TC08	User fills all fields correctly and taps Save	Form validates successfully and user is navigated to Home Screen.	pass

Sprint Test Case1



Sprint Test Case2

Sprint 1 / Test Cases – Add Budget

Test Case No.	Test Case Description	Expected Results	Outcome (Pass / Fail / Comments)
TC09	User taps the ✓ (check) button with the Amount field empty	A SnackBar message appears: “Enter a valid amount”.	pass
TC10	User enters a non-numeric amount (e.g., “ss”) and taps ✓	A SnackBar message appears: “Enter a valid amount”.	pass
TC11	User enters a valid amount and selects a category (e.g., “Transport”), then taps ✓	A new document is added in the commitments collection with category, amount, notes, and createdAt. Profile income is decreased using FieldValue.increment(-amount). SnackBar displays: “Added and total income updated ”.	pass
TC12	User enters a valid amount but does not select any category, then taps ✓	Commitment is added with category = "Other". Firestore updates the profile income. SnackBar confirms success.	pass
TC13	User opens the screen with existing commitments	A list of commitments is displayed, sorted by createdAt (descending). Each entry shows category, amount, and notes (or “No notes” if empty).	pass
TC14	User deletes a commitment from the list	The selected commitment document is removed from Firestore. The deleted amount is added back to the total income. The row disappears immediately.	pass

Sprint Test Case2

Coding

EXPLORER

- M
  - android
    - settings.gradle.kts
    - build
    - ios
    - lib
      - Screens
        - addBudget_screen.dart 1
        - home_screen.dart 1
        - login_screen.dart 1
        - personal_information_screen.dart 2
        - sign_up_screen.dart
        - splash_screen.dart
      - main.dart 1
    - linux
    - macos
    - test
    - web
    - windows
    - .flutter-plugins-dependencies
    - .gitignore
    - .metadata
    - analysis_options.yaml

main.dart 1 × | Flutter (Pixel 8) | pubspec.yar

lib > main.dart > MaliApp

```dart
import 'package:flutter/material.dart';
import 'package:google_fonts/google_fonts.dart';
import 'package:firebase_core/firebase_core.dart';
import 'Screens/splash_screen.dart';
import 'Screens/login_screen.dart';

Run | Debug | Profile
Future<void> main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();

  final app = Firebase.app();
  debugPrint('✅ Firebase app: ${app.name}');
  runApp(const MaliApp());
}

class MaliApp extends StatelessWidget {
  const MaliApp({super.key});

  @override
  Widget build(BuildContext context) {
    final theme = ThemeData(
      useMaterial3: true,
      colorScheme: ColorScheme.fromSeed(
        seedColor: const Color(0xFF3E6E50),
        primary: const Color(0xFF3E6E50),
```

EXPLORER

- M
  - android
    - settings.gradle.kts
    - build
    - ios
    - lib
      - Screens
        - addBudget_screen.dart 1
        - home_screen.dart 1
        - login_screen.dart 1
        - personal_information_screen.dart 2
        - sign_up_screen.dart
        - splash_screen.dart
      - main.dart 1
    - linux
    - macos
    - test
    - web
    - windows
    - .flutter-plugins-dependencies
    - .gitignore
    - .metadata
    - analysis_options.yaml

build.gradle.kts ...\app | Flutter (Pixel 8)

lib > Screens > splash_screen.dart > _SplashBackground > build

```dart
import 'dart:async';
import 'package:flutter/material.dart';
import 'package:google_fonts/google_fonts.dart';
import 'login_screen.dart';

class splash_screen extends StatefulWidget {
  const splash_screen({super.key});

  @override
  State<splash_screen> createState() => _SplashScreenState();
}

class _SplashScreenState extends State<splash_screen> {
  @override
  void initState() {
    super.initState();

    Timer(const Duration(seconds: 4), () {
      if (!mounted) return;
      Navigator.pushReplacement(
        context,
        MaterialPageRoute(builder: (_) => const LoginPage()),
      );
    }); // Timer
  }
```

EXPLORER

M

- android
- settings.gradle.kts
- build
- ios
- lib
  - Screens
    - addBudget_screen.dart 1
    - home_screen.dart 1
    - login_screen.dart 1
    - personal_information_screen.dart 2
    - sign_up_screen.dart
    - splash_screen.dart
  - main.dart 1
- linux
- macos

login_screen.dart 1 | Flutter (Pixel 8)

lib > Screens > home_screen.dart > ...

```dart
import 'package:flutter/material.dart';
import 'package:google_fonts/google_fonts.dart';
import 'package:cloud_firestore/cloud_firestore.dart';

import 'addBudget_screen.dart';
import 'personal_information_screen.dart';

class HomeScreen extends StatefulWidget {
  const HomeScreen({super.key});
  @override
  State<HomeScreen> createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
  int _currentIndex = 0;

  @override
  Widget build(BuildContext context) {
```

EXPLORER

MALI

- m
  - .dart_tool
  - .idea
  - android
  - build
  - ios
  - lib
    - Screens
      - addBudget_screen.dart 1
      - home_screen.dart 1
      - login_screen.dart 1
      - my_daily_expenses_screen.dart
      - personal_information_screen.dart 2
      - saving_goals_screen.dart
      - sign_up_screen.dart
      - splash_screen.dart
    - main.dart 1
  - linux
  - macos
  - test
  - web
  - windows

OUTLINE

TIMELINE

DEPENDENCIES

personal_information_screen.dart 2 | login_screen.dart 1 | addBudget_screen.dart 1 | home_screen.dart 1

m > lib > Screens > personal_informatio

```dart
import 'package:flutter/material.dart';
import 'package:google_fonts/google_fonts.dart';
import 'package:cloud_firestore/cloud_firestore.dart';

import 'home_screen.dart';

class PersonalInformationScreen extends StatefulWidget {
  final String? prefilledName;
  final String? prefilledEmail;

  final bool isNewUser;

  const PersonalInformationScreen({
    super.key,
    this.prefilledName,
    this.prefilledEmail,
    this.isNewUser = false,
  });

  @override
  State<PersonalInformationScreen> createState() =>
      _PersonalInformationScreenState();
}

class _PersonalInformationScreenState extends State<PersonalInformationScreen> {
  final _formKey = GlobalKey<FormState>();

  final _name = TextEditingController();
  final _email = TextEditingController();
```

M

- android
- settings.gradle.kts
- build
- ios
- lib
  - Screens
    - addBudget_screen.dart 1
    - home_screen.dart 1
    - login_screen.dart 1
    - personal_information_screen.dart 2
    - sign_up_screen.dart
    - splash_screen.dart
  - main.dart 1
- linux
- macos
- test
- web
- windows

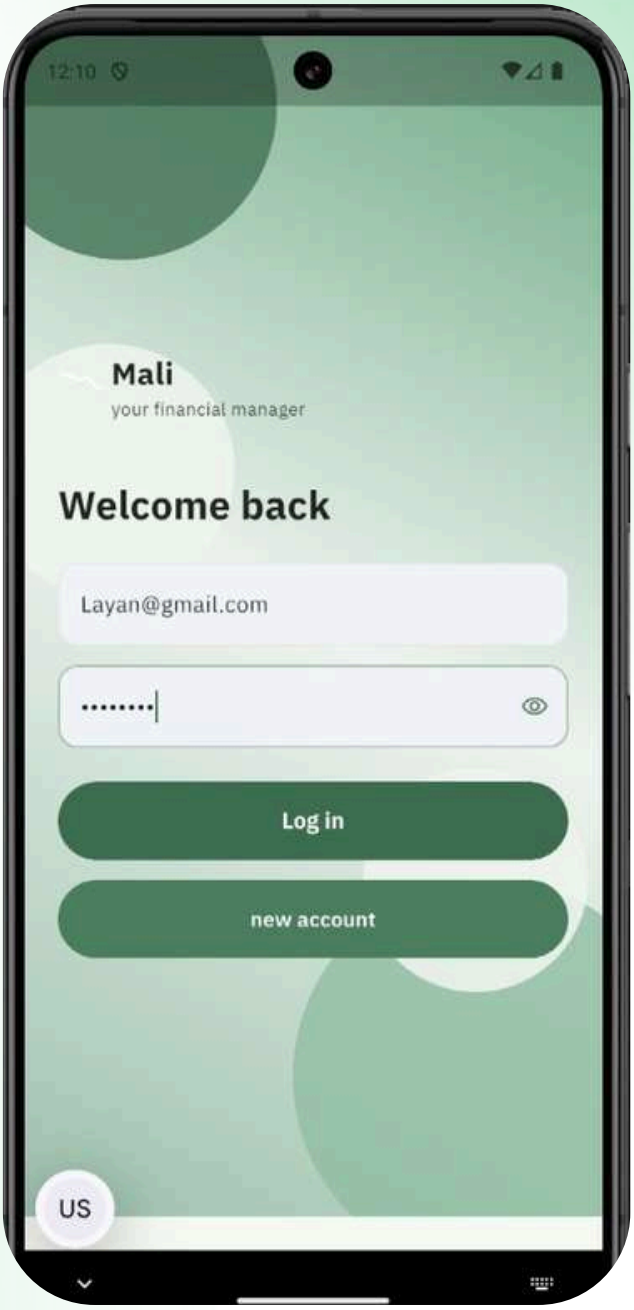
Flutter (Pixel 8)

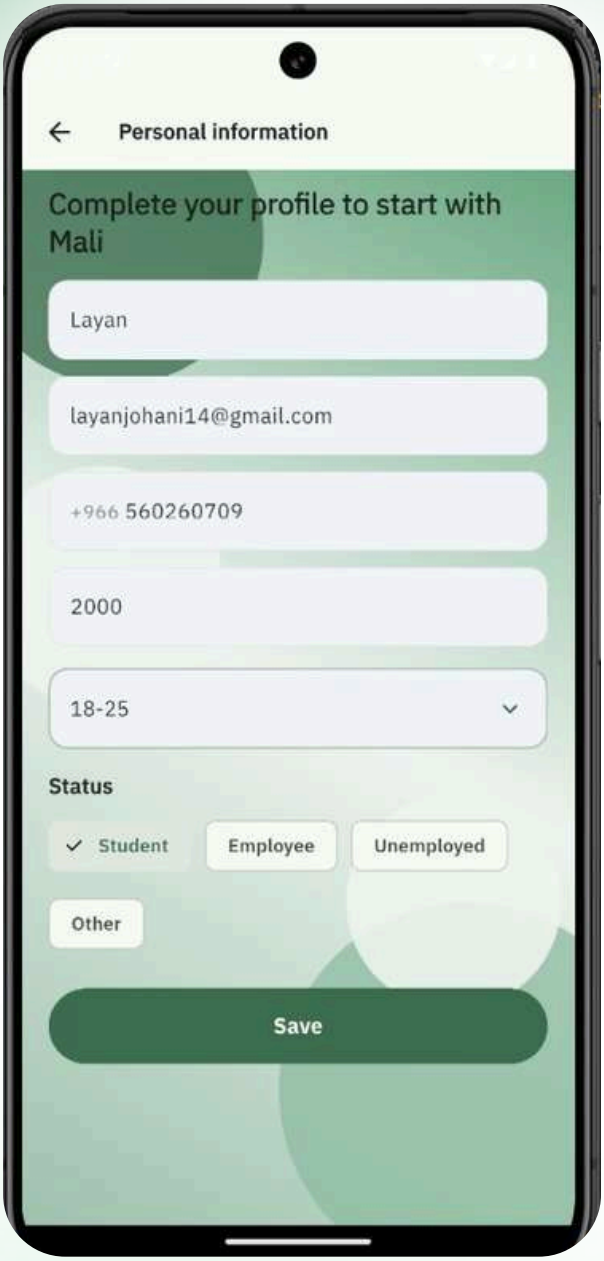
lib > Screens > addBudget_screen.dart > _AddBudgetScreenState > _categories

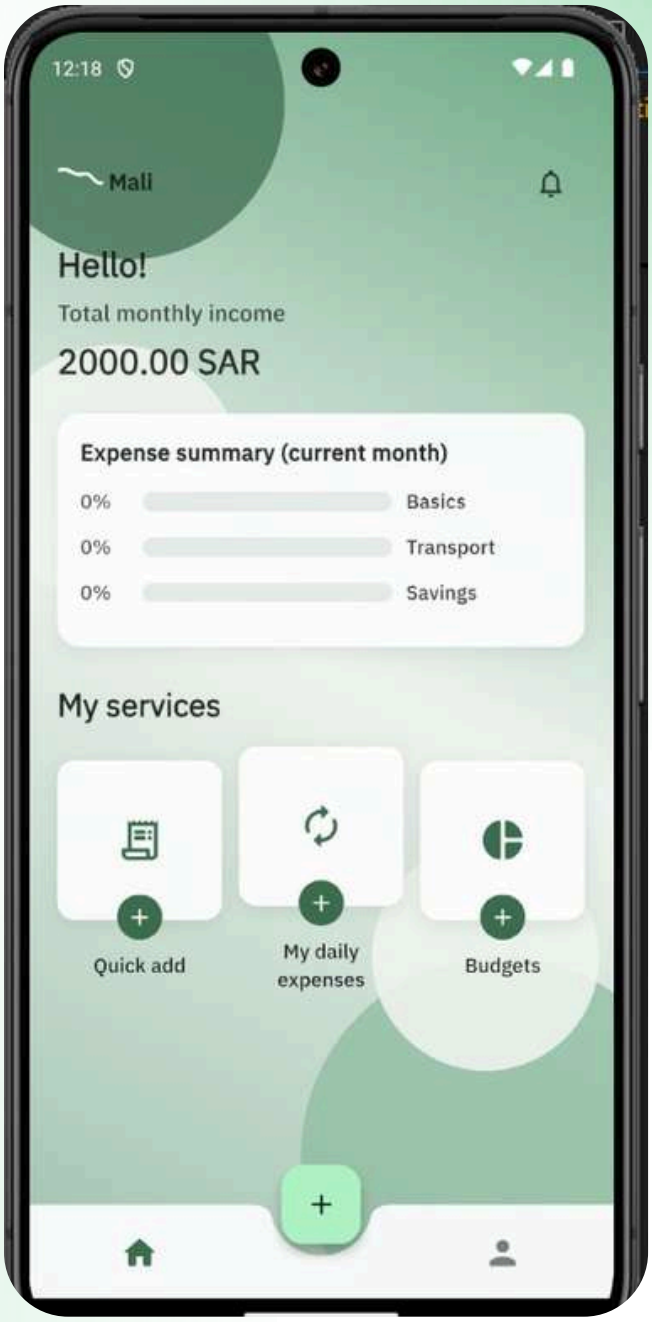
```dart
import 'package:flutter/material.dart';
import 'package:google_fonts/google_fonts.dart';
import 'package:cloud_firestore/cloud_firestore.dart';

import 'home_screen.dart';
import 'personal_information_screen.dart';

class addBudget_screen extends StatefulWidget {
  const addBudget_screen({super.key});
  @override
  State<addBudget_screen> createState() => _AddBudgetScreenState();
}

class _AddBudgetScreenState extends State<addBudget_screen> {
  final _amountCtrl = TextEditingController();
  final _notesCtrl = TextEditingController();
  String? _selectedCategory;

  //Firestore
  final _commitmentsCol = FirebaseFirestore.instance.collection('commitments');
  final _profileDoc = FirebaseFirestore.instance
```

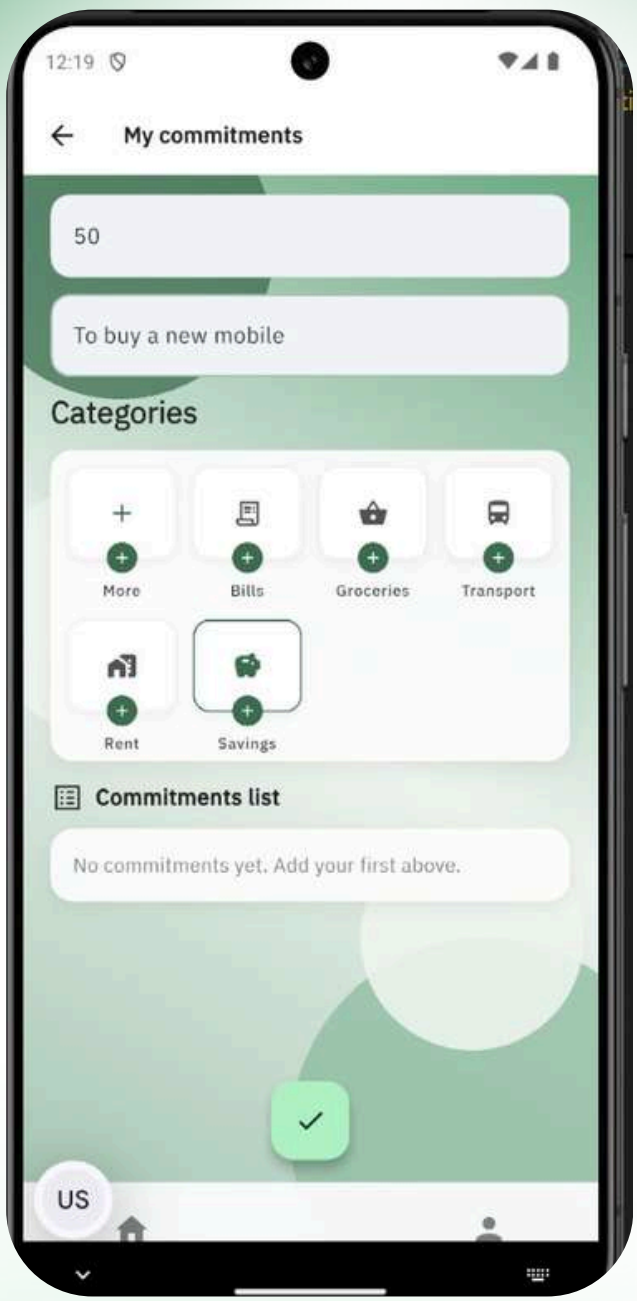
Interfaces

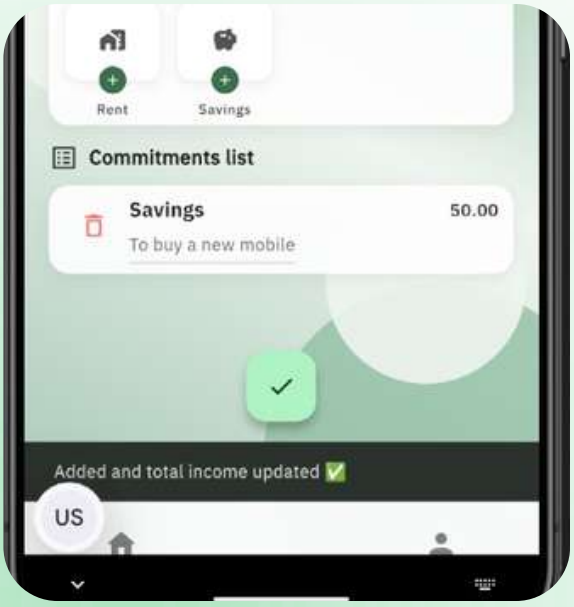
11:42
Mali
Your smart financial assistant

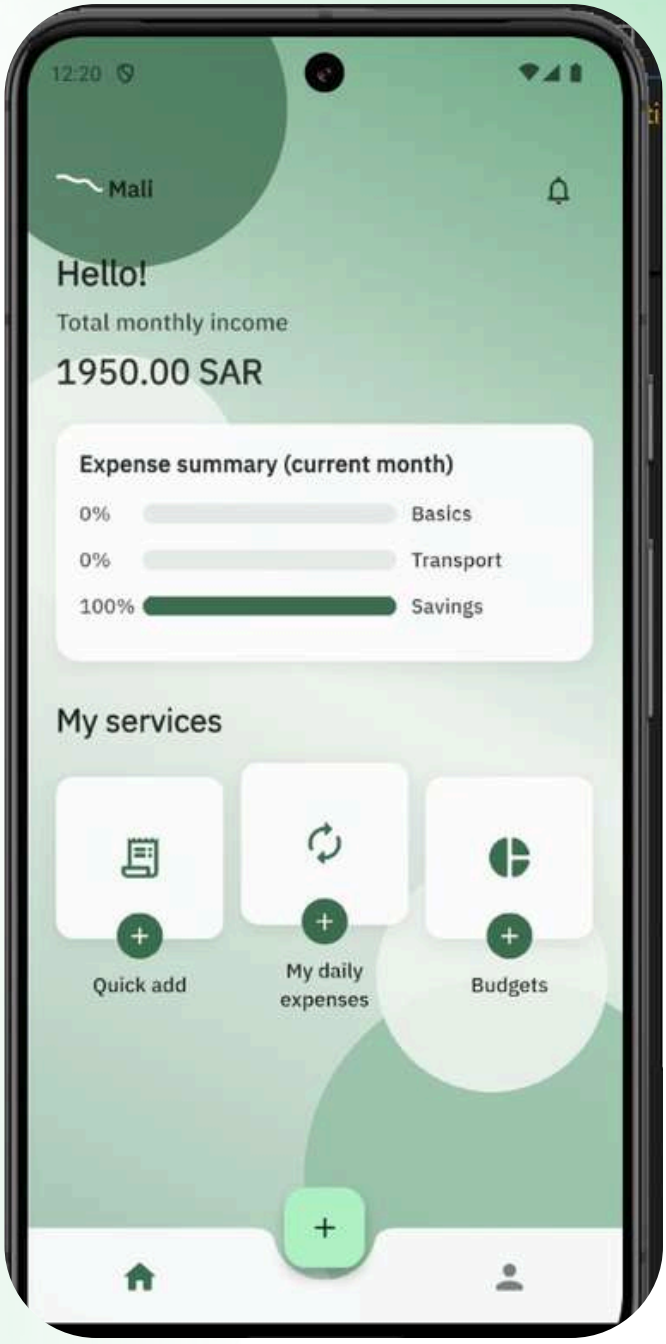

12:10
Mali
your financial manager
Welcome back
Layan@gmail.com
Log in
new account
US


Personal information
Complete your profile to start with Mali
Layan
layanjohani14@gmail.com
+966 560260709
2000
18-25
Status
Student
Employee
Unemployed
Other
Save


12:18
Mali
Hello!
Total monthly income
2000.00 SAR
Expense summary (current month)
0% Basics
0% Transport
0% Savings
My services
Quick add
My daily expenses
Budgets

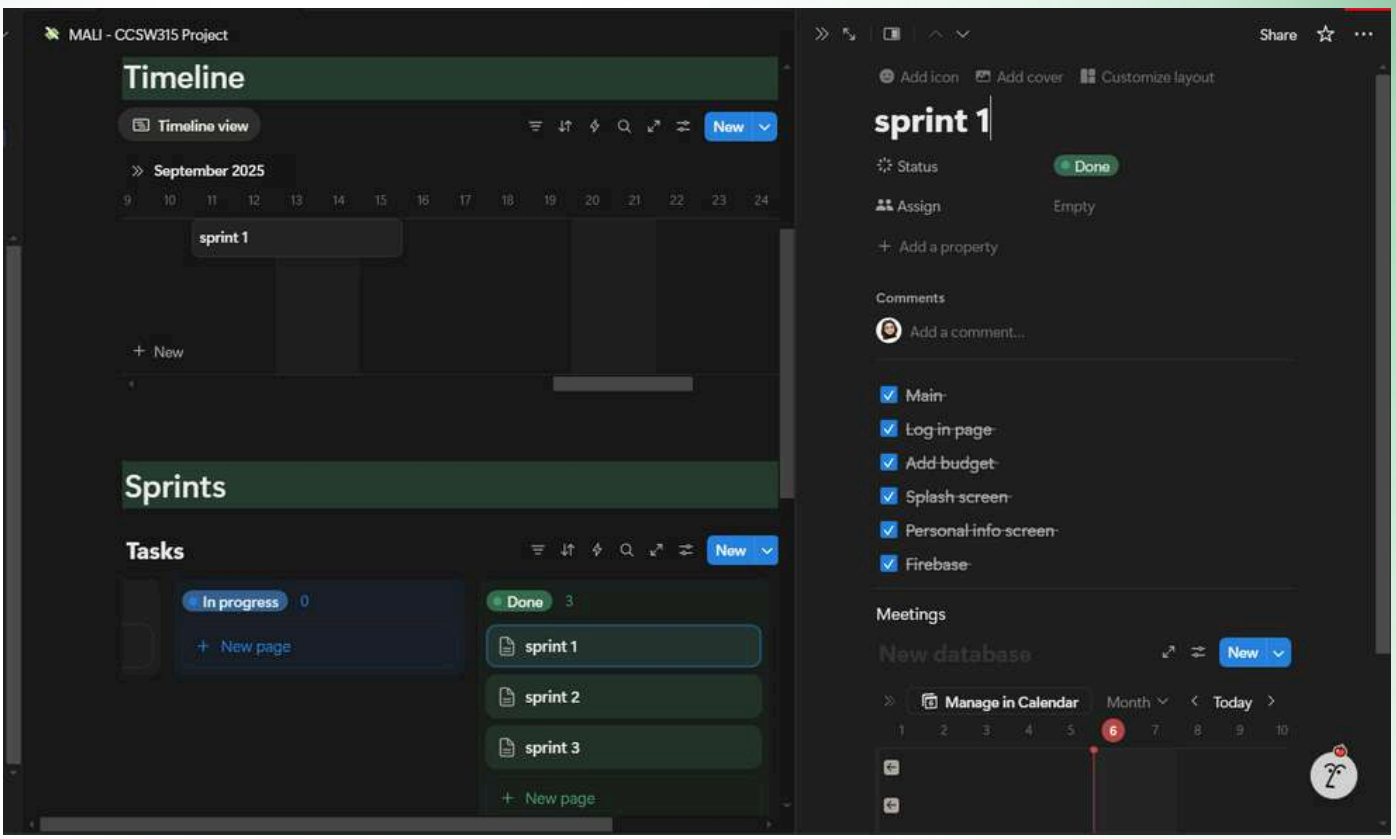

12:19
My commitments
50
To buy a new mobile
Categories
More
Bills
Groceries
Transport
Rent
Savings
Commitments list
No commitments yet. Add your first above.
US


Rent
Savings
Commitments list
Savings
50.00
To buy a new mobile
Added and total income updated ✅
US

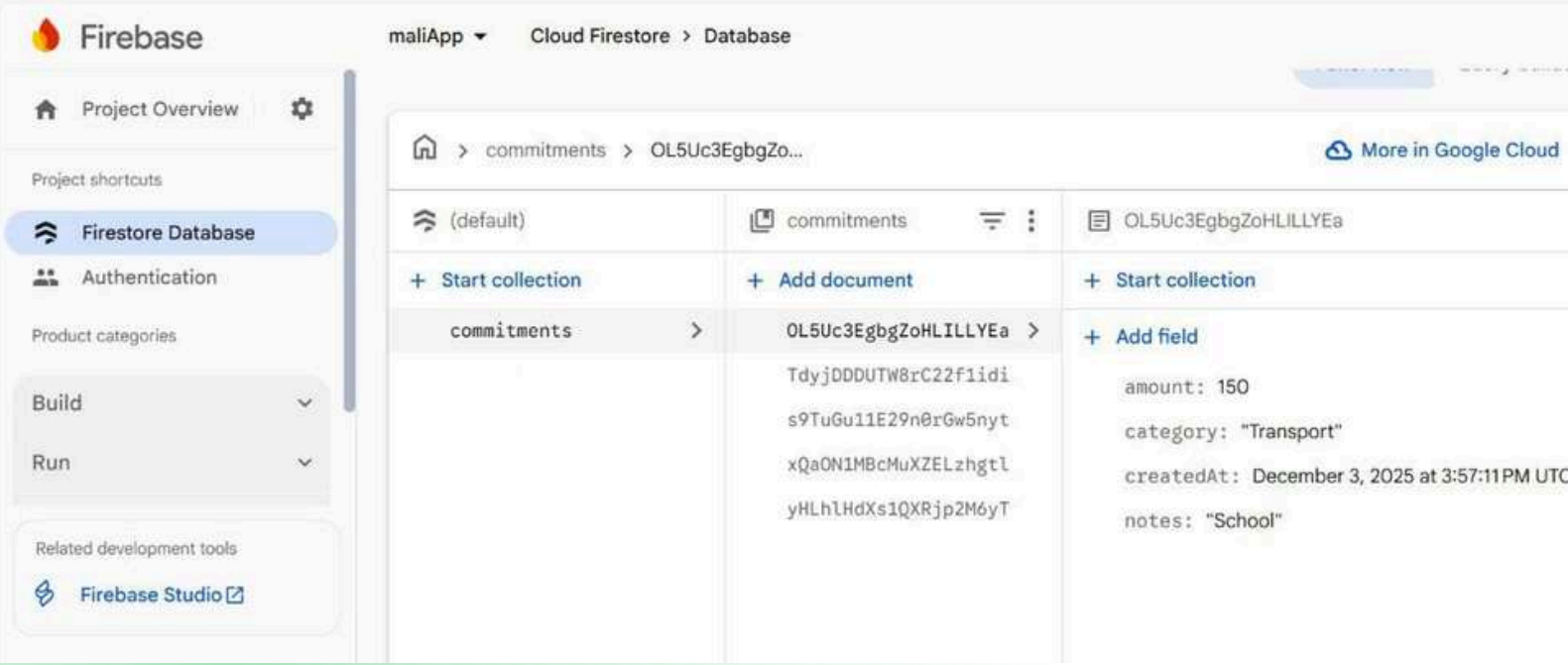

12:20
Mali
Hello!
Total monthly income
1950.00 SAR
Expense summary (current month)
0% Basics
0% Transport
100% Savings
My services
Quick add
My daily expenses
Budgets

Tools

Notion



Firebase



Sprint 2

Project Initiation

Project Initiation

Project Name: Mali App

Project Members: Joury Almutairi – Hanan Alnejaimi – Reema Alzahrani – Layan Aljohani – Elaf Althobaiti

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Project Description:

Mali is a budgeting application designed to help users manage their income and track daily expenses.
In Sprint 2, the focus was on implementing the Daily Expenses feature, allowing users to view all expenses recorded for the current day, filter expenses by category, and display the total spent amount for that day in real-time.

Prioritized Functional and non-functional requirements:

Functional Requirements (FR-2):

The user shall be able to enter their monthly income and track daily expenses . the system shall update and display the remaining balance accordingly

Non-Functional Requirements:

1- NFR-1: Usability - The screen must be simple and easy to understand
2- NFR-2: Performance Searching and listing expenses must be fast even with large data.
3- NFR-3: Reliability**:** Real-time data updates must work consistently

EXAMPLE IF YOU CHOOSE SCRUM: Story Backlog :

Component Name: Daily Expenses

Story Name: View Today's Expenses
Story Sequence No: 001

Story Short Description:
Users can view today's recorded expenses in real time.

Project Initiation

Story Long Description:
The system retrieves expenses created within today's date range using Firestore. The expenses are sorted by timestamp and displayed in a scrollable list, including category, amount, notes, and time. The data updates in real time to reflect new entries or changes.

Component Name: search

Story Name: Category Search
Story Sequence No: 002

Story Short Description:
Users can search for expenses by entering a category name.

Story Long Description:
When the user types in the search field, the system filters the list of expenses based on case-insensitive and partial text matching. If no matches are found, the system displays a message notifying the user.

Prioritize stories and define sprints:

Sprint #1:

- Personal Information screen

- Home screen with income summary

- Budget allocation & category distribution

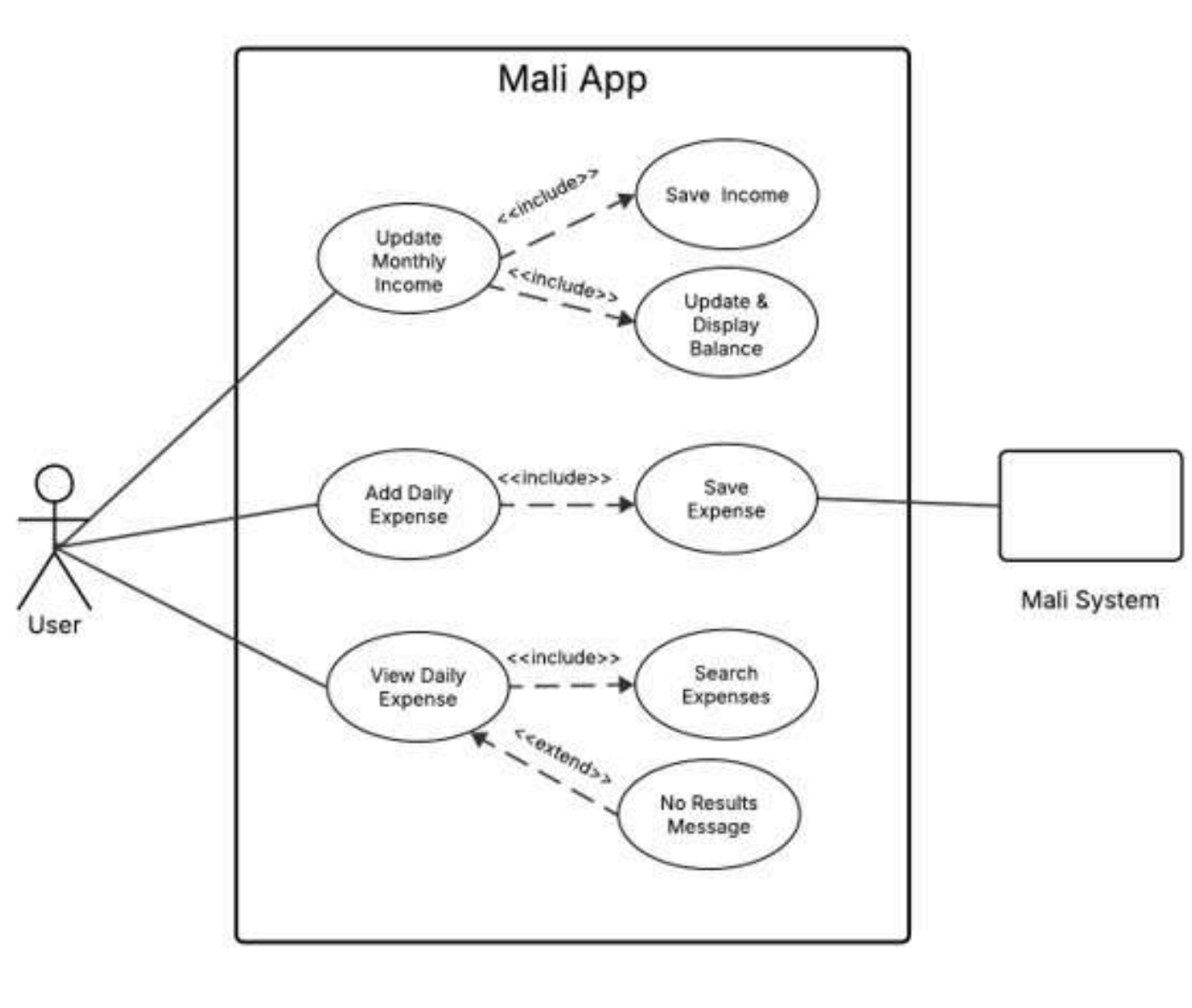
- Basic data validation and saving

-Initial test cases

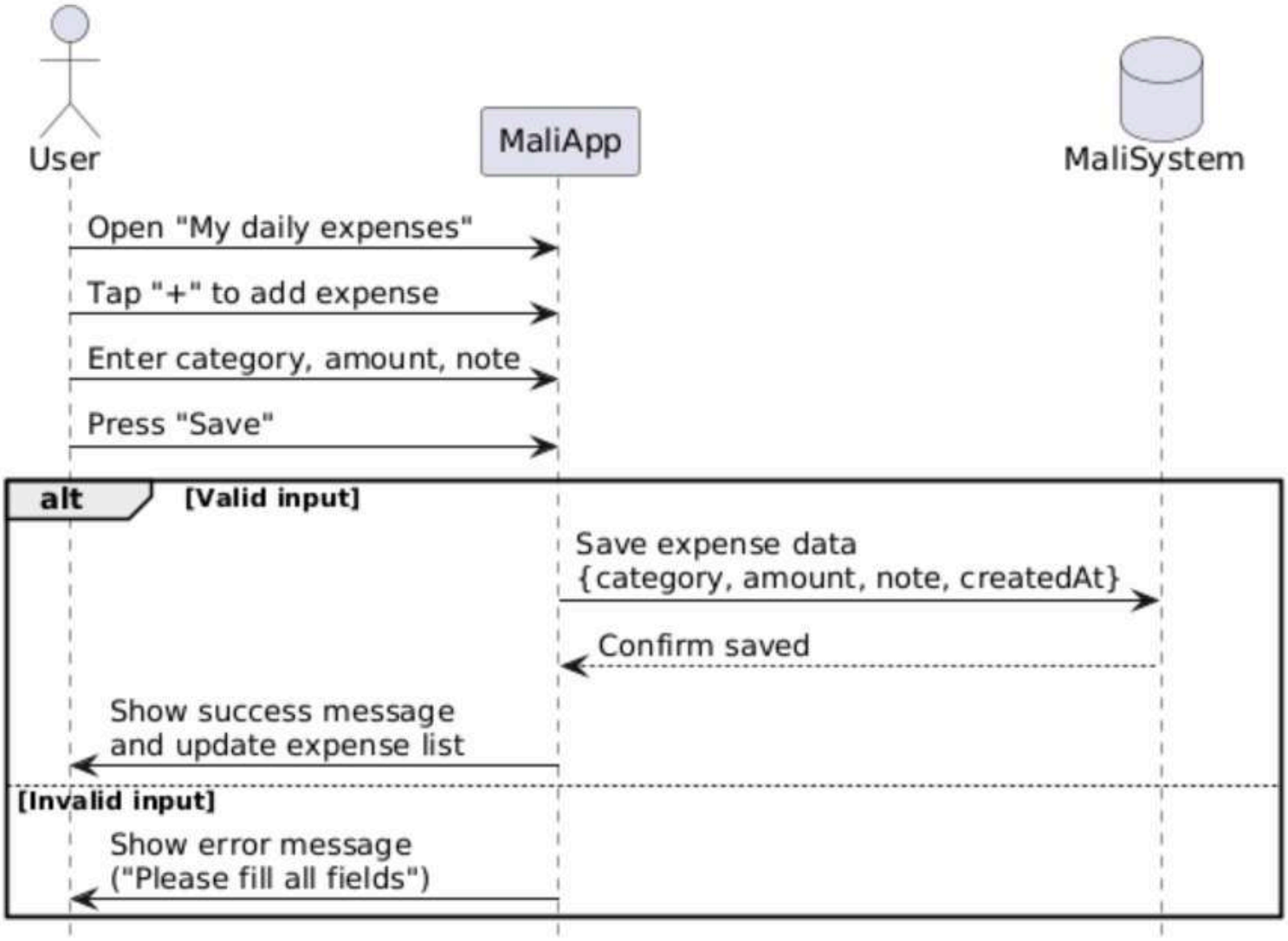
Sprint #2:

- Daily Expenses screen (My Daily Expenses)
- Real-time expense listing via Firestore
- Search and filter by category
- Dynamic calculation of Total Spent Today
- Empty state messages
- Add expense shortcut via "+" button
- Full testing for filtering, totals, and performance

✦ Usecase


Mali App
Update Monthly Income
<<include>>
Save Income
<<include>>
Update & Display Balance
Add Daily Expense
<<include>>
Save Expense
View Daily Expense
<<include>>
Search Expenses
<<extend>>
No Results Message
User
Mali System

✦ Sequence Diagram



Sprint Meeting(1)

Sprint Meeting(s)

Project Name: Mali

Project Members: Hanan Alnejaimi - Reema Alzahrani - Layan Aljohani - Joury Almutairi - Elaf Althubaiti

---

Sprint #2 Stand up Meeting - [8/11/2025]

Sprint Duration: 4 weeks

Scrum Master: Dr. Malak Alharbi

Client: Dr. Malak Alharbi

Pair Programmers: Layan Aljohani & Reema Alzahrani

Stories:

Component Name	Story Sequence Number	Use Cases (e.g., functionalities)
Daily expenses	001	Users can track daily expenses
Search Bar	002	Users can search expenses by category
Total spent balance	003	Users can view the spent balance for the same day

Follow-up meeting questions:

1. **What has been completed since the last meeting?** We reviewed all of Sprint 2 requirements and identified the key areas that need improvement and further focus.
2. **What are you going to be working on next?** We will begin designing the Daily Expenses interface screens and proceed with the coding.
3. **Do you have any issues/impediments?** No

Scrum's Master comments based on the above questions:

Sprint Meeting(2)

Sprint Meeting(s)

Project Name: Mali

Project Members: Hanan Alnejaimi - Reema Alzahrani - Layan Aljohani - Joury Almutairi - Elaf Althubaiti

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Sprint #2 Stand up Meeting - [10/11/2025]

Sprint Duration: 4 weeks

Scrum Master: Dr. Malak Alharbi

Client: Dr. Malak Alharbi

Pair Programmers: Layan Aljohani & Reema Alzahrani

Stories:

Component Name	Story Sequence Number	Use Cases (e.g., functionalities)
Daily expenses	001	Users can track daily expenses
Search Bar	002	Users can search expenses by category
Total spent balance	003	Users can view the spent balance for the same day

Follow-up meeting questions:

1. **What has been completed since the last meeting?** We completed daily expenses interface screens
2. **What are you going to be working on next?** We'll move on to connecting the interface with Firebase so user expenses can be stored and accessed properly.
3. **Do you have any issues/impediments?** No

Scrum's Master comments based on the above questions:

Sprint Meeting(3)

Sprint Meeting(s)

Project Name: Mali

Project Members: Hanan Alnejaimi - Reema Alzahrani - Layan Aljohani - Joury Almutairi - Elaf Althubaiti

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Sprint #2 Stand up Meeting - [11/11/2025]

Sprint Duration: 4 weeks

Scrum Master: Dr. Malak Alharbi

Client: Dr. Malak Alharbi

Pair Programmers: Layan Aljohani & Reema Alzahrani

Stories:

Component Name	Story Sequence Number	Use Cases (e.g., functionalities)
Daily expenses	001	Users can track daily expenses
Search Bar	002	Users can search expenses by category
Total spent balance	003	Users can view the spent balance for the same day

Follow-up meeting questions:

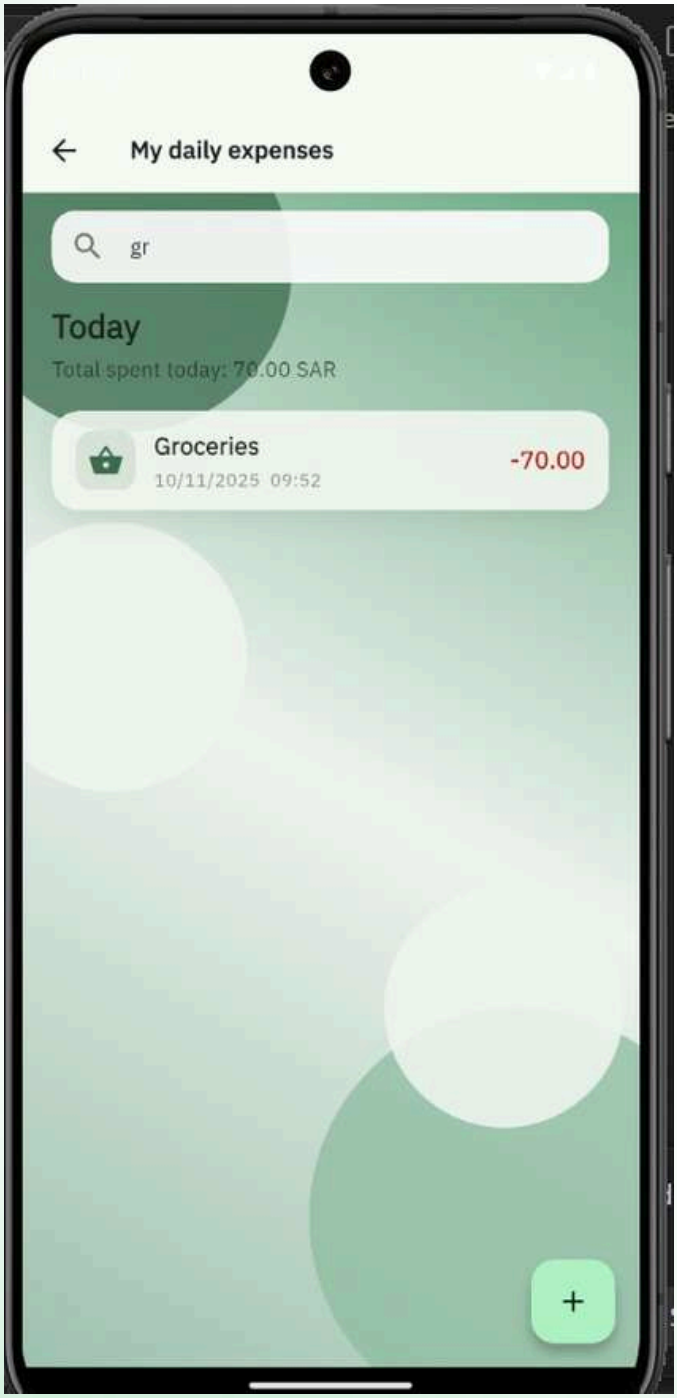
1. **What has been completed since the last meeting?** We implemented the backend integration, connecting the Daily Expenses module to Firebase
2. **What are you going to be working on next?** We will start preparing for Sprint 3
3. **Do you have any issues/impediments?** No

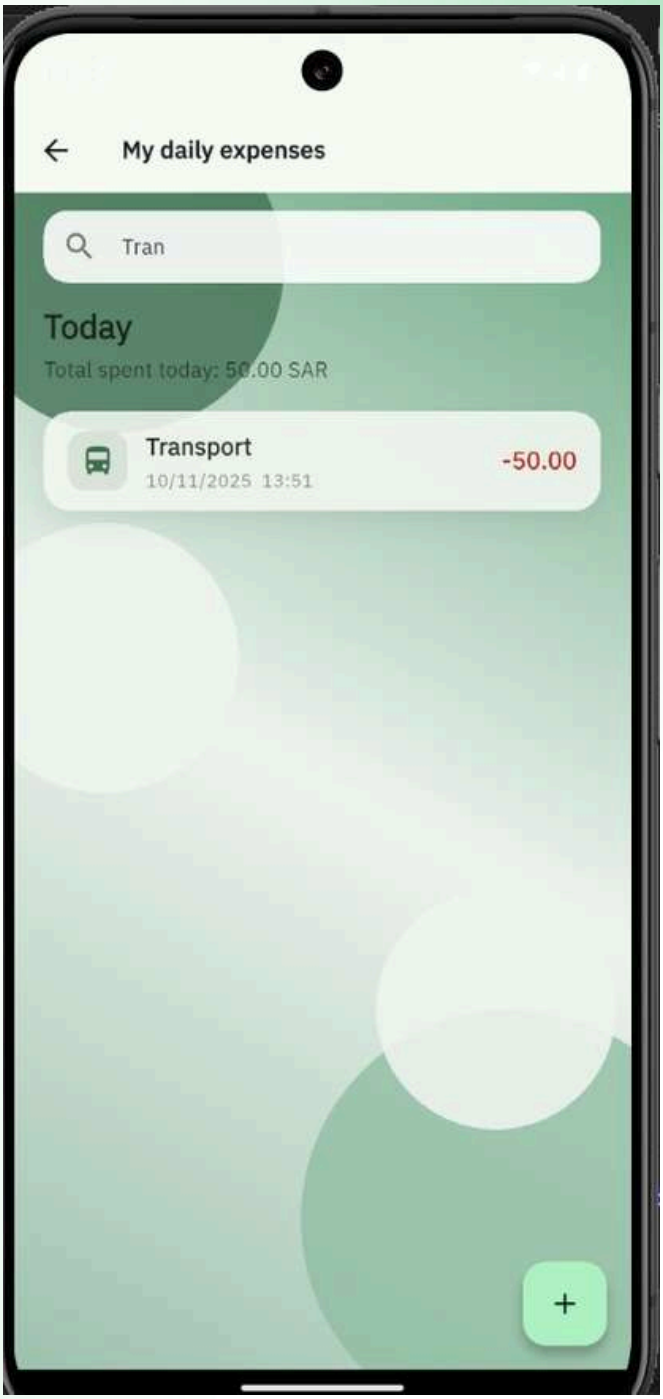
Scrum's Master comments based on the above questions:

Sprint Test Case

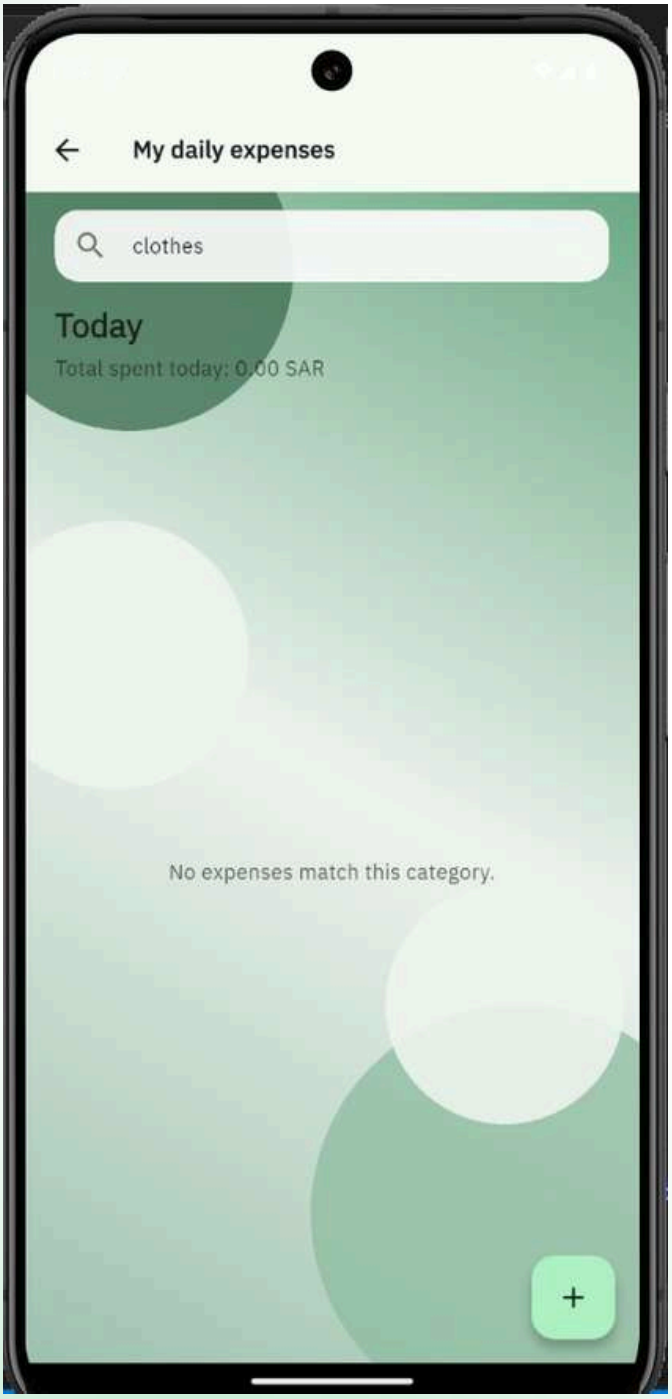
Test Case No.	Test Case Description	Expected Results	Outcome (Pass / Fail / Comments)
TC01	User searches exact category name (e.g., "Groceries")	Only expenses belonging to Groceries appear. The displayed Total spent today updates based on filtered list only.	Pass
TC02	User searches by partial text (e.g., "Tran")	Only expenses belonging to categories that contain the substring (e.g., *Transport*) appear. Search should be case-insensitive.	Pass
TC03	User searches for a non-existing category (e.g., "Clothes")	No expense cards should be displayed. A message appears: "No expenses match this category."	Pass
TC04	User searches category in different letter cases (e.g., "savings" vs "SAVINGS")	Same results appear in both cases (search is case-insensitive).	Pass
TC05	Validate Total Calculation	Displayed Total spent today equals the sum of the amounts of the visible filtered expenses only.	Pass

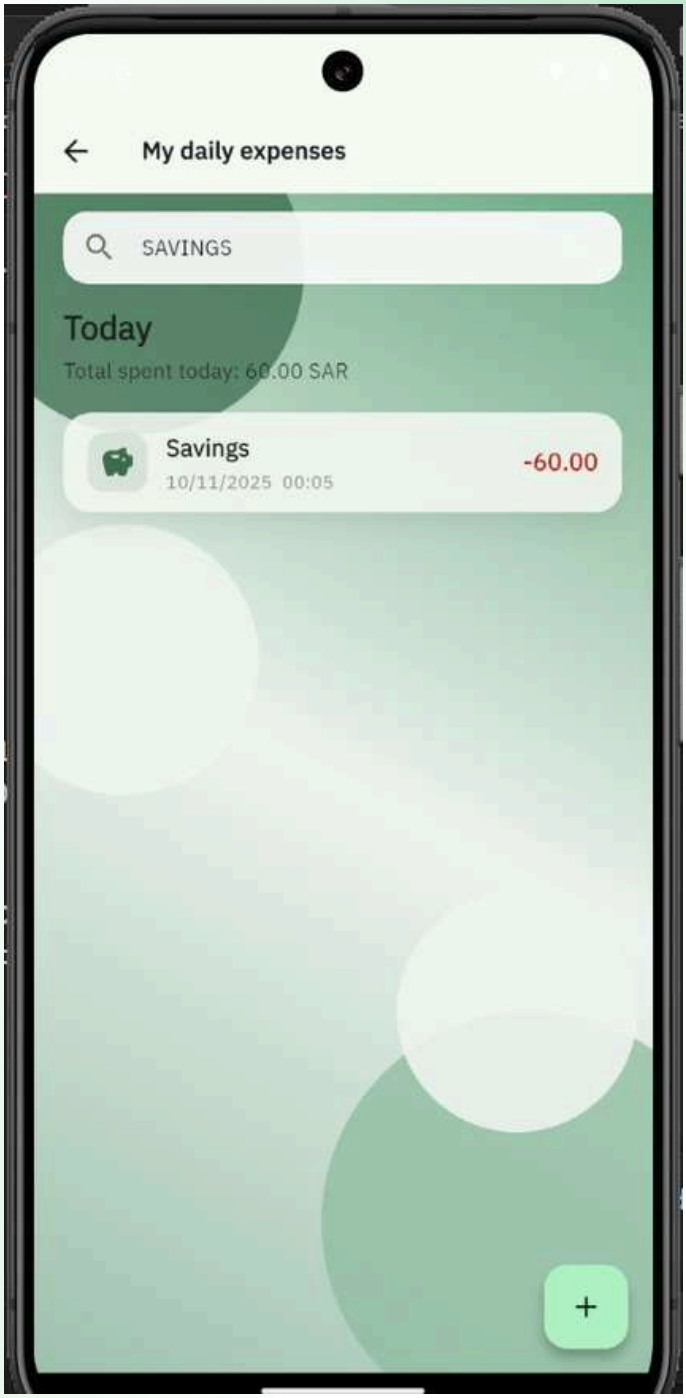
Sprint Test Case





Sprint Test Case





Coding

EXPLORER

- M
  - android
    - gradlew.bat
    - local.properties
    - m_android.iml
    - settings.gradle.kts
  - build
  - ios
  - lib
    - Screens
      - addBudget_screen.dart 1
      - home_screen.dart 1
      - login_screen.dart 1
      - my_daily_expenses_screen.dart
      - personal_information_screen.dart 2
      - saving_goals_screen.dart
      - sign_up_screen.dart
      - splash_screen.dart
    - main.dart 1
  - linux
  - macos
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      - sign_up_screen.dart
      - splash_screen.dart
    - main.dart 1
  - linux
  - macos
  - test
  - web
  - windows
- OUTLINE
- TIMELINE
- DEPENDENCIES

build.gradle.kts android | splash_screen.dart | home_screen (3).dart | sign_up_screen.da

lib > Screens > home_screen.dart > ...

```dart
import 'package:flutter/material.dart';
import 'package:google_fonts/google_fonts.dart';
import 'package:cloud_firestore/cloud_firestore.dart';

import 'addBudget_screen.dart';
import 'personal_information_screen.dart';
import 'my_daily_expenses_screen.dart';
import 'saving_goals_screen.dart';

class HomeScreen extends StatefulWidget {
  const HomeScreen({super.key});
  @override
  State<HomeScreen> createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
  int _currentIndex = 0;

  @override
  Widget build(BuildContext context) {
    final cs = Theme.of(context).colorScheme;
    final text = GoogleFonts.ibmPlexSansArabicTextTheme(
      Theme.of(context).textTheme,
    );

    return Scaffold(
      extendBody: true,
      body: Stack(
        children: [
          const _HomeSoftGreenBackground(),
          SafeArea(
            child: SingleChildScrollView(
              padding: const EdgeInsets.fromLTRB(20, 32, 20, 120),
              child: Column(
                crossAxisAlignment: CrossAxisAlignment.start,
                children: [
                  Row(
                    children: [
                      const _HomeLogoMark(),
                      const SizedBox(width: 10),
                      Text(
                        'Mali',
                        style: text.titleMedium!.copyWith(
                          fontWeight: FontWeight.w700,
                          color: cs.onSurface.withOpacity(.95),
                        ),
                      ), // Text
                      const Spacer(),
                      IconButton(
                        icon: const Icon(Icons.notifications_none_rounded),
                        onPressed: () {},
                        color: cs.onSurface.withOpacity(.8),
```

Coding

EXPLORER

- M
  - android
    - gradlew.bat
    - local.properties
    - m_android.iml
    - settings.gradle.kts
  - build
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- lib
  - Screens
    - addBudget_screen.dart 1
    - home_screen.dart 1
    - login_screen.dart 1
    - my_daily_expenses_screen.dart
    - personal_information_screen.dart 2
    - saving_goals_screen.dart
    - sign_up_screen.dart
    - splash_screen.dart
  - main.dart 1
- linux
- macos
- test
- web
- windows
- OUTLINE
- TIMELINE

- DEPENDENCIES

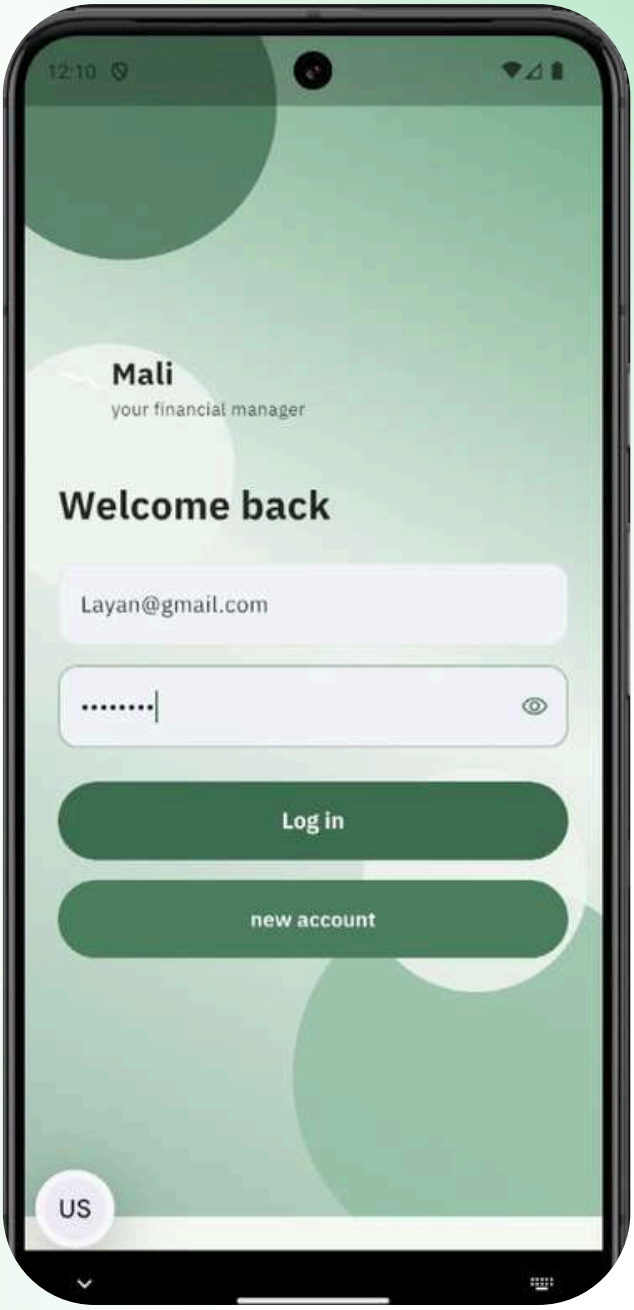
my_daily_expenses_screen.dart × | google-services.json | build.gradle.kts ...\app

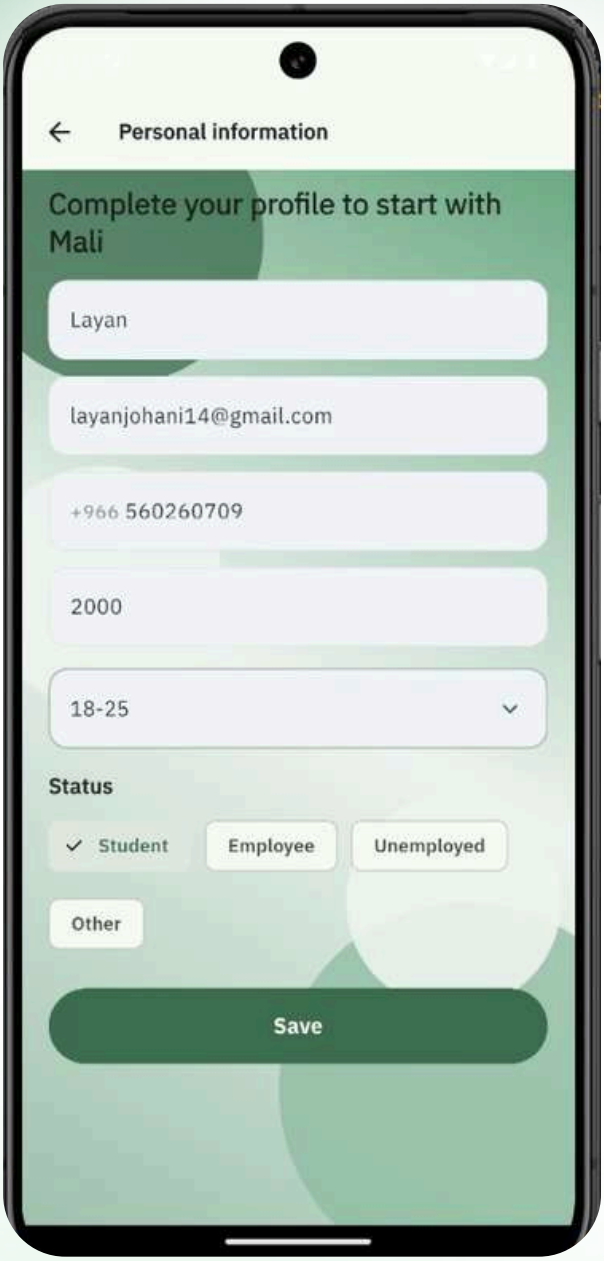
lib > Screens > my_daily_expenses_screen > dispose

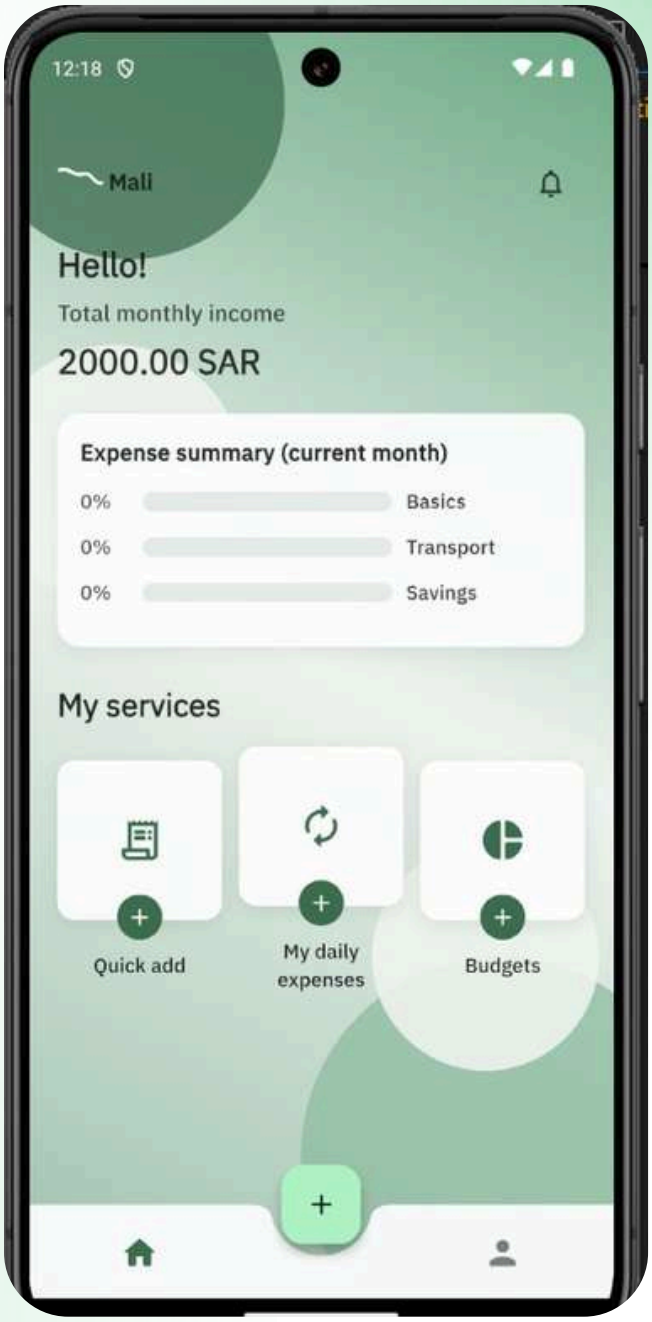
```
1  import 'package:flutter/material.dart';
2  import 'package:google_fonts/google_fonts.dart';
3  import 'package:cloud_firestore/cloud_firestore.dart';
4
5  import 'addBudget_screen.dart';
6
7  class MyDailyExpensesScreen extends StatefulWidget {
8    const MyDailyExpensesScreen({super.key});
9
10   @override
11   State<MyDailyExpensesScreen> createState() => _MyDailyExpensesScr
12 }
13
14 class _MyDailyExpensesScreenState extends State<MyDailyExpensesScre
15   final TextEditingController _searchController = TextEditingContro
16   String _searchCategory = '';
17
18   @override
19   void dispose() {
20     _searchController.dispose();
21     super.dispose();
22   }
23
24   @override
25   Widget build(BuildContext context) {
26     final cs = Theme.of(context).colorScheme;
27     final text = GoogleFonts.ibmPlexSansArabicTextTheme(
28       Theme.of(context).textTheme,
30
31     final now = DateTime.now();
32     final dayStart = DateTime(now.year, now.month, now.day);
33     final dayEnd = dayStart.add(const Duration(days: 1));
34
35     return Scaffold(
36       extendBody: true,
37       appBar: AppBar(
38         backgroundColor: Colors.transparent,
39         elevation: 0,
40         foregroundColor: cs.onSurface,
41         title: Text(
42           'My daily expenses',
43           style: text.titleMedium!.copyWith(fontWeight: FontWeight.w70
44         ), // Text
45       ), // AppBar
46       body: Stack(
47         children: [
48           const _SoftGreenBackgroundDE(),
49           SafeArea(
50             child: Padding(
51               padding: const EdgeInsets.fromLTRB(20, 12, 20, 20),
52               child: StreamBuilder<QuerySnapshot<Map<String, dynamic>>
53                 stream: FirebaseFirestore.instance
54                     .collection('commitments')
55                     .where('createdAt', isGreaterThanOrEqualTo: daySta
56                     .where('createdAt', isLessThan: dayEnd)
```

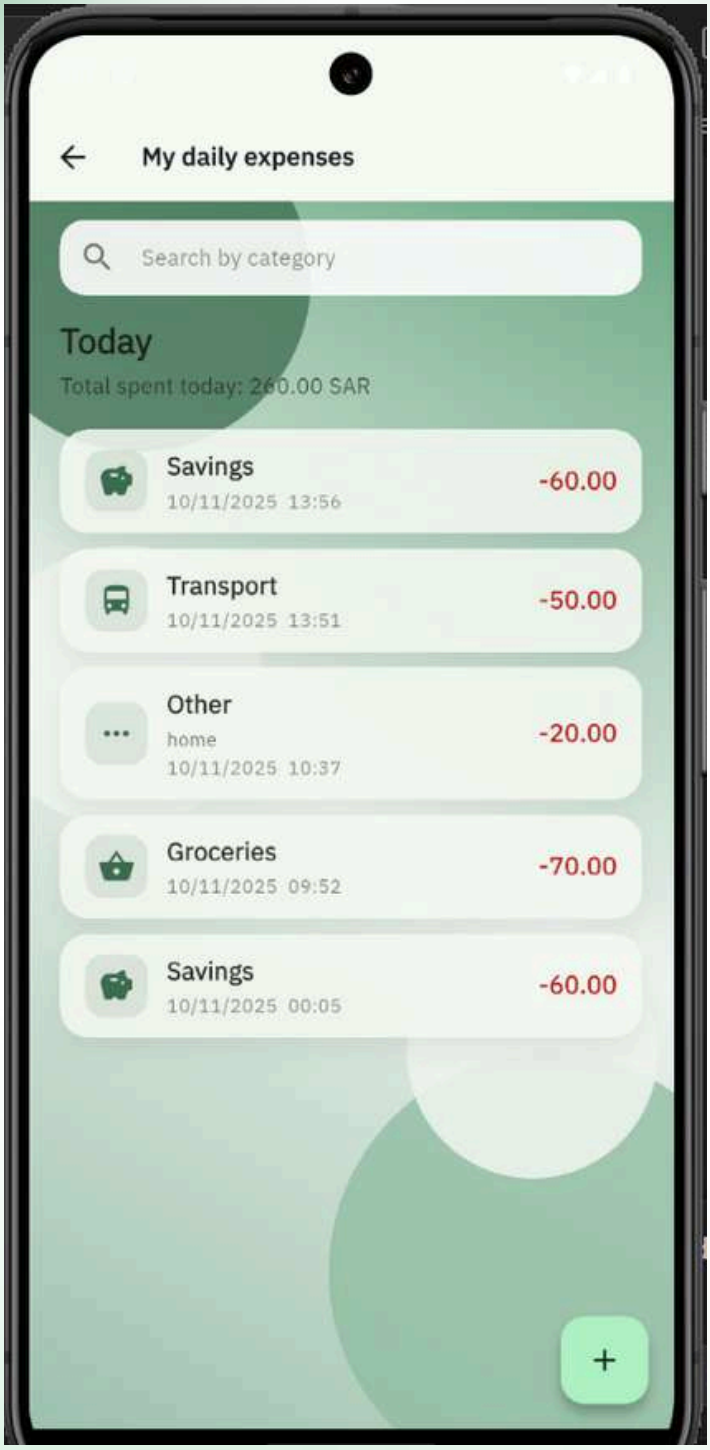
Interfaces

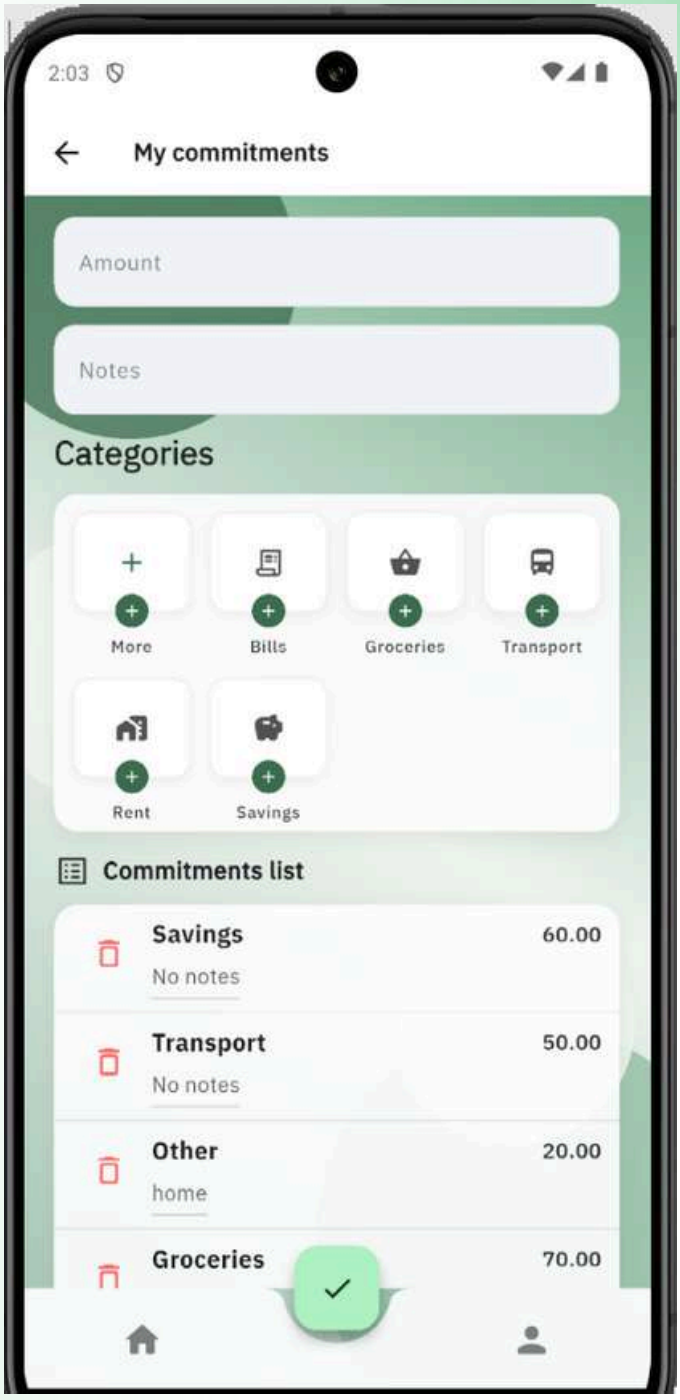
11:42
Mali
Your smart financial assistant

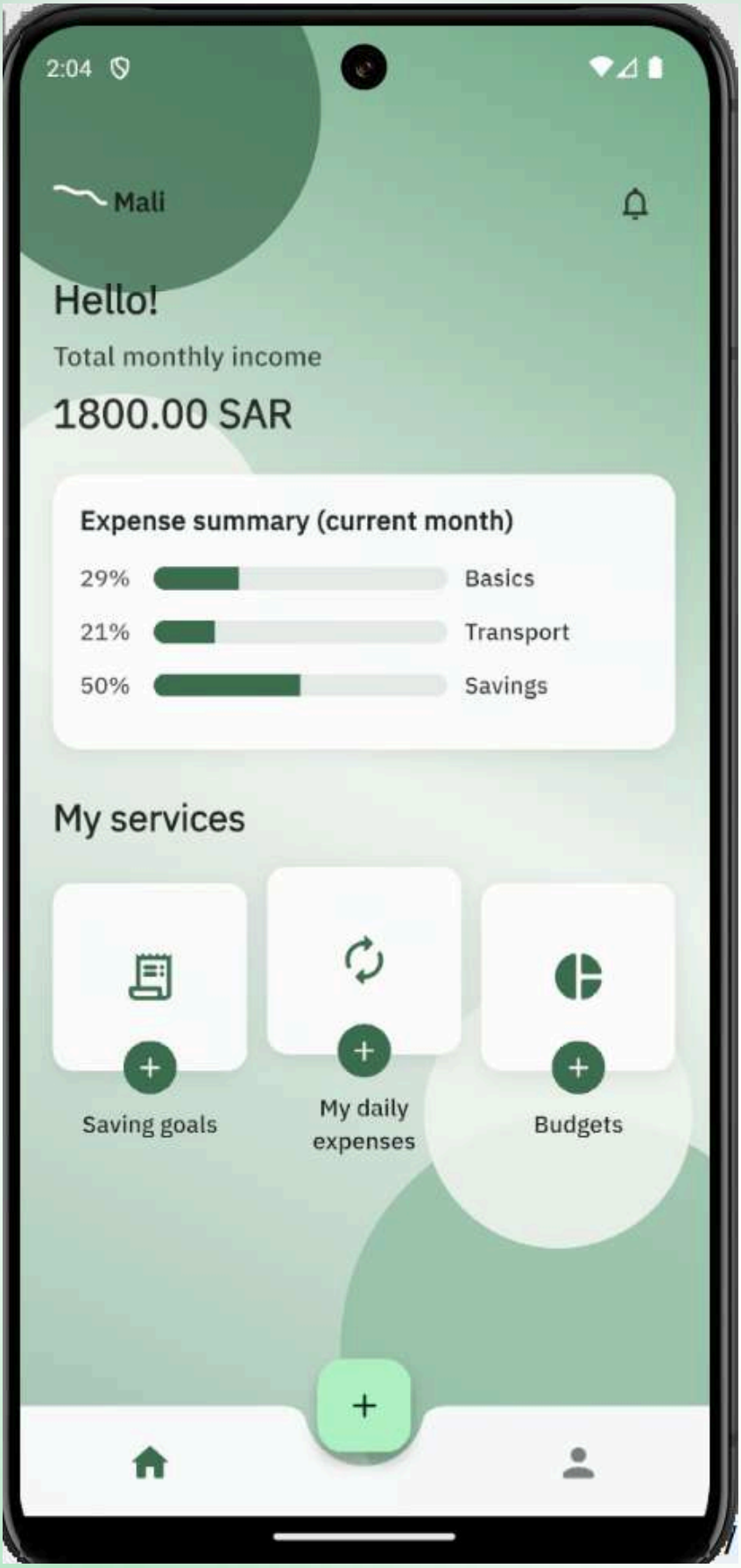

12:10
Mali
your financial manager
Welcome back
Layan@gmail.com
Log in
new account
US


Personal information
Complete your profile to start with Mali
Layan
layanjohani14@gmail.com
+966 560260709
2000
18-25
Status
Student
Employee
Unemployed
Other
Save


12:18
Mali
Hello!
Total monthly income
2000.00 SAR
Expense summary (current month)
0% Basics
0% Transport
0% Savings
My services
Quick add
My daily expenses
Budgets

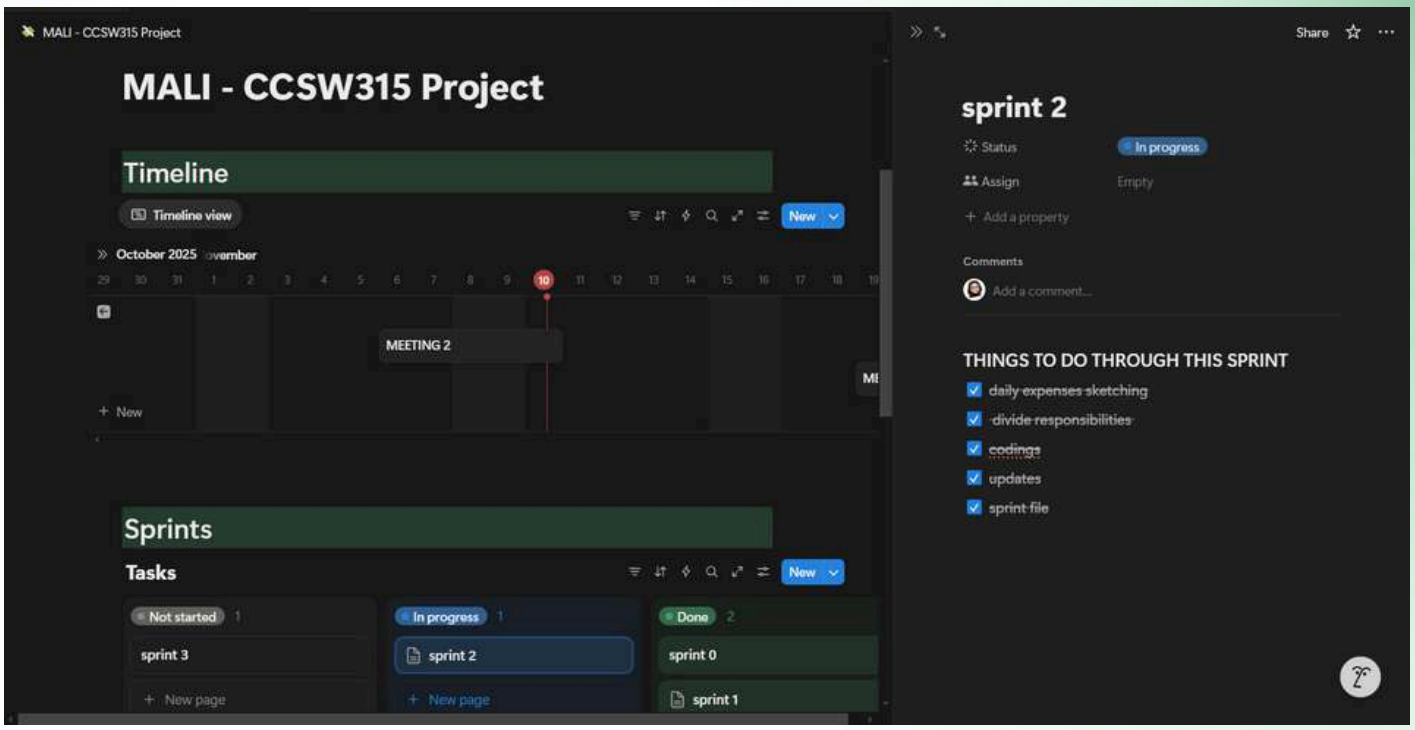

My daily expenses
Search by category
Today
Total spent today: 260.00 SAR
Savings
10/11/2025 13:56
-60.00
Transport
10/11/2025 13:51
-50.00
Other
home
10/11/2025 10:37
-20.00
Groceries
10/11/2025 09:52
-70.00
Savings
10/11/2025 00:05
-60.00


2:03
My commitments
Amount
Notes
Categories
More
Bills
Groceries
Transport
Rent
Savings
Commitments list
Savings
No notes
60.00
Transport
No notes
50.00
Other
home
20.00
Groceries
70.00

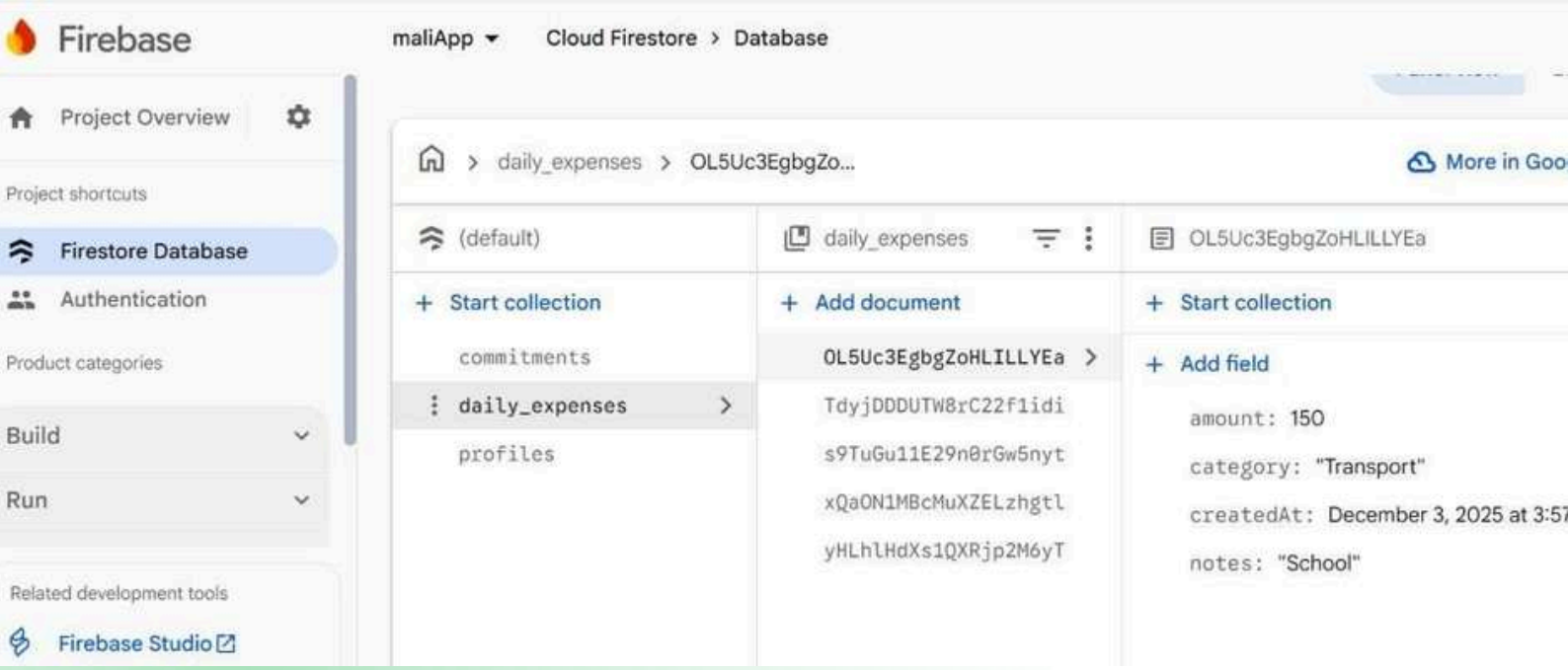

2:04
Mali
Hello!
Total monthly income
1800.00 SAR
Expense summary (current month)
29%
Basics
21%
Transport
50%
Savings
My services
Saving goals
My daily expenses
Budgets

Tools

Notion



Firebase



Sprint 3

Project Initiation

Project Initiation

Project Name:

Mali App

Project Members:

Joury Almutairi – Hanan Alnejaimi – Reema Alzahrani – Layan Aljohani – Elaf Althobaiti

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Project Description:

Mali is a budgeting application designed to help users manage their income and expenses.
In Sprint 3, the focus was on implementing the Saving Goals feature, which allows users to:

- Add new saving goals
- Enter total price and saved amount
- Validate incorrect input
- Track progress visually
- Update saved amount
- Automatically detect when a goal is completed

This feature helps users stay motivated and monitor their progress toward their financial goals.

Prioritized Functional and non-functional requirements:

Functional Requirement (FR-3):

The user shall be able to set saving goals and track their progress towards achieving these goals.

Non-Functional Requirements:

1. Usability: Interface must be simple and easy to understand.
2. Performance: Updating saved amounts and recalculating progress must be fast.
3. Reliability: Progress tracking must always be accurate.

Project Initiation

SCRUM Story Backlog

Component Name: Saving Goals

Story Name: Add Saving Goal
Story Sequence No: 001

Story Short Description:
Users can add a new saving goal including name, total price, and saved amount.

Story Long Description:
Users enter goal details in a dialog. Validation ensures required fields are filled, total price is positive, and saved amount is not greater than total or negative. When valid, a new goal card appears showing progress and remaining amount.

Component Name: Saving Goals

Story Name: Update Saved Amount
Story Sequence No: 002

Story Short Description:
Users can update the saved amount for an existing goal.

Story Long Description:
When the edit icon is tapped, the user can update the saved amount. The system validates the amount, recalculates progress, and displays a completion message when the saved amount reaches the total price.

Prioritize Stories and Define Sprints

Sprint #1:

- Personal Information screen
- Home screen with income summary
- Budget allocation & category distribution
- Basic data validation and saving
- Initial test cases

Sprint #2:

- Daily Expenses screen
- Real-time expense listing using Firestore
- Search and filter by category

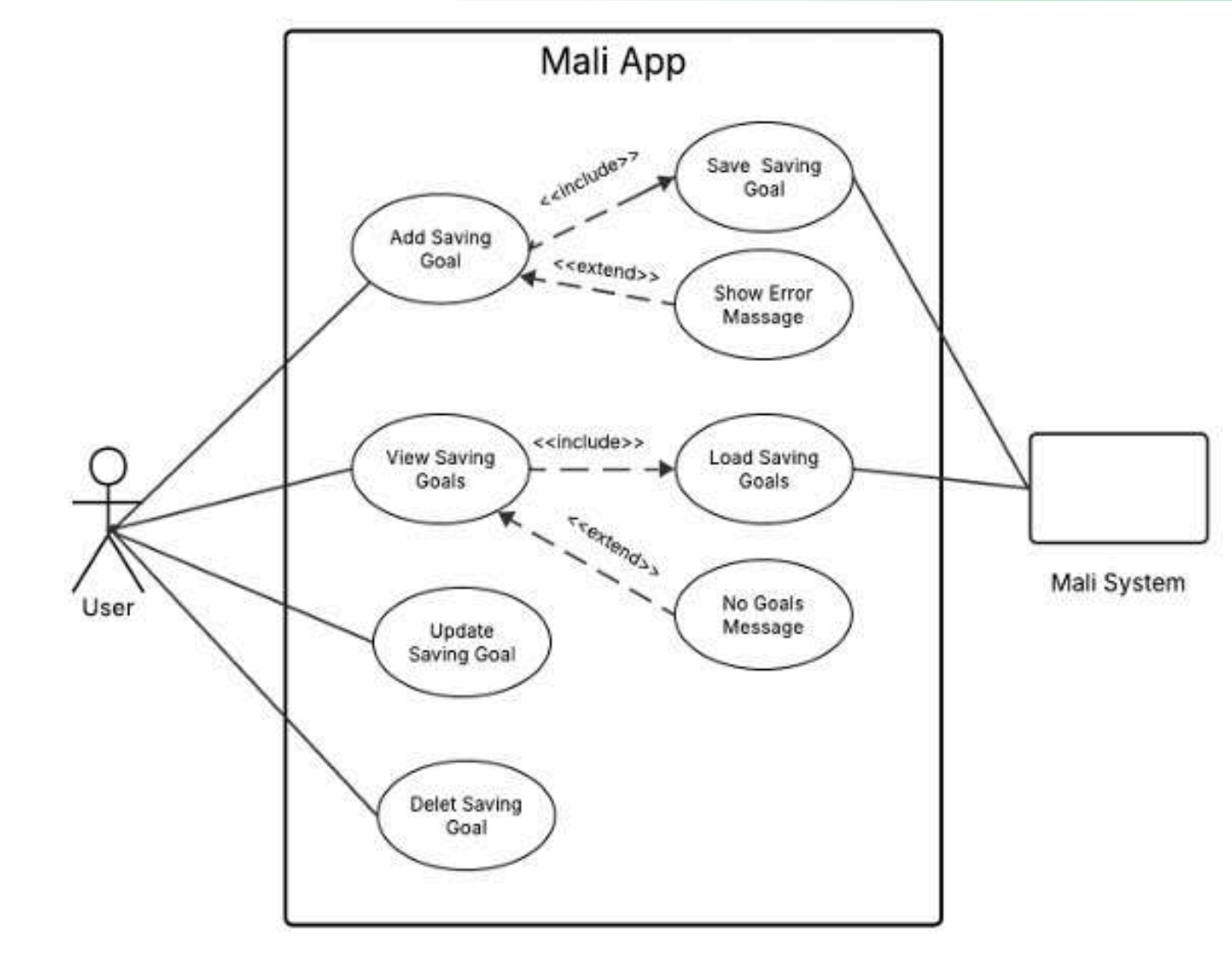
Project Initiation

- Dynamic calculation of today's total spent
- Empty state messages
- Add expense via “+” button
- Full testing for filtering and totals

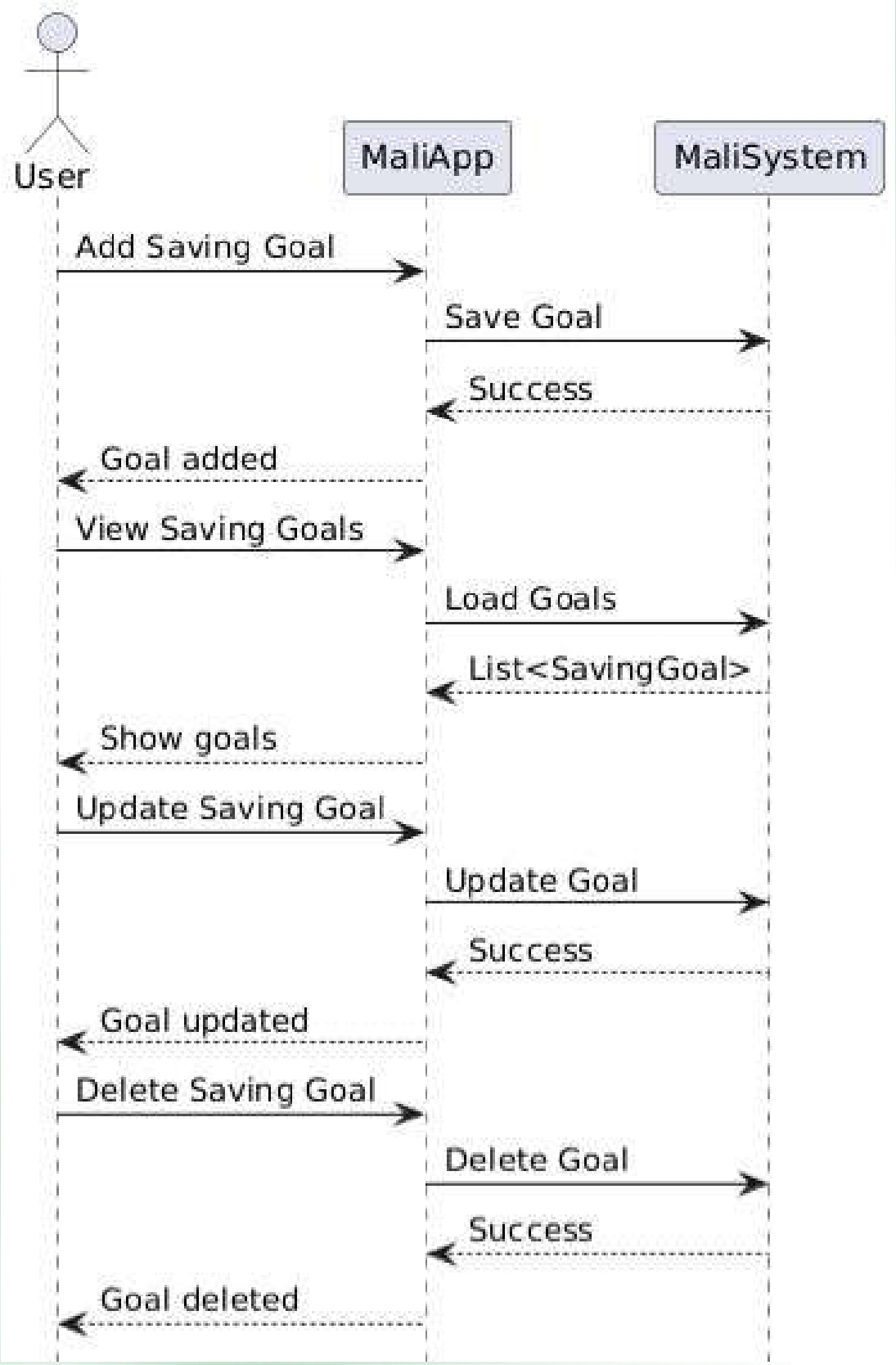
Sprint #3:

- Saving Goals screen
- Add new saving goal
- Validation for goal input
- Progress tracking and dynamic update
- Edit saved amount dialog
- Completion status message
- Full testing for saving goals feature

✦ Usecase


Mali App
Save Saving Goal
<<include>>
Add Saving Goal
<<extend>>
Show Error Massage
View Saving Goals
<<include>>
Load Saving Goals
<<extend>>
No Goals Message
Update Saving Goal
Delet Saving Goal
User
Mali System

Sequence Diagram



Sprint Meeting(1)

Sprint Meeting(s)

Project Name: Mali

Project Members: Reema Alzahrani – Layan Aljohani – Joury Almutairi – Hanan Alnejaimi – Elaf althubaiti

---

Sprint #*i* Stand up Meeting - [2025-11-24]

Sprint Duration: **3 weeks**

Scrum Master: **Dr. Malak Alharbi**

Client: **Dr. Malak Alharbi**

Pair Programmers: **Layan Aljohani & Reema Alzahrani**

Stories:

Component Name	Story Sequence Number	Use Cases (e.g., functionalities)
Saving goals	004	**User can create saving goals and track progress.**
Summary dashboard	005	System displays income, expenses , savings, and remaining balance.

Follow-up meeting questions :

1. What has been completed since the last meeting?
   Completed UI polishing for main screens – connected saving goals to database
2. What are you going to be working on next?
   Testing user flow between screens - Fixing minor UI responsiveness issues.
3. Do you have any issues/impediments? No

Scrum's Master comments based on the above questions:

Sprint Meeting(2)

Sprint Meeting(s)

Project Name: Mali

Project Members: Reema Alzahrani – Layan Aljohani – Joury Almutairi – Hanan Alnejaimi – Elaf althubaiti

---

Sprint #*i* Stand up Meeting - [27-11-2025]

Sprint Duration: 3 weeks

Scrum Master: Dr. Malak Alharbi

Client: Dr. Malak Alharbi

Pair Programmers: Layan Aljohani & Reema Alzahrani

Stories:

Component Name	Story Sequence Number	Use Cases (e.g., functionalities)
Final Review & Testing	006	Ensure all features (Saving Goals + Dashboard) work correctly with no errors.

Follow-up meeting questions :

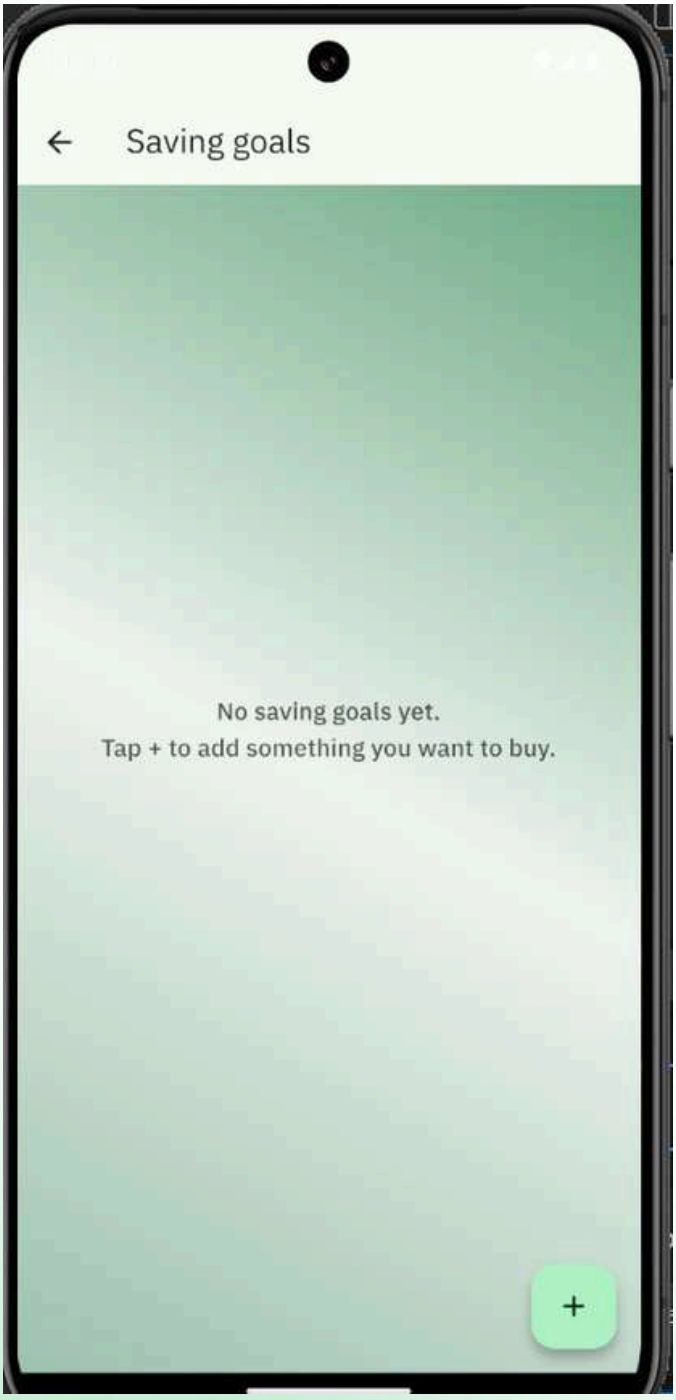
1. What has been completed since the last meeting?
   Verified dashboard calculations
2. What are you going to be working on next?
   Preparing for the sprint demo
3. Do you have any issues/impediments? No

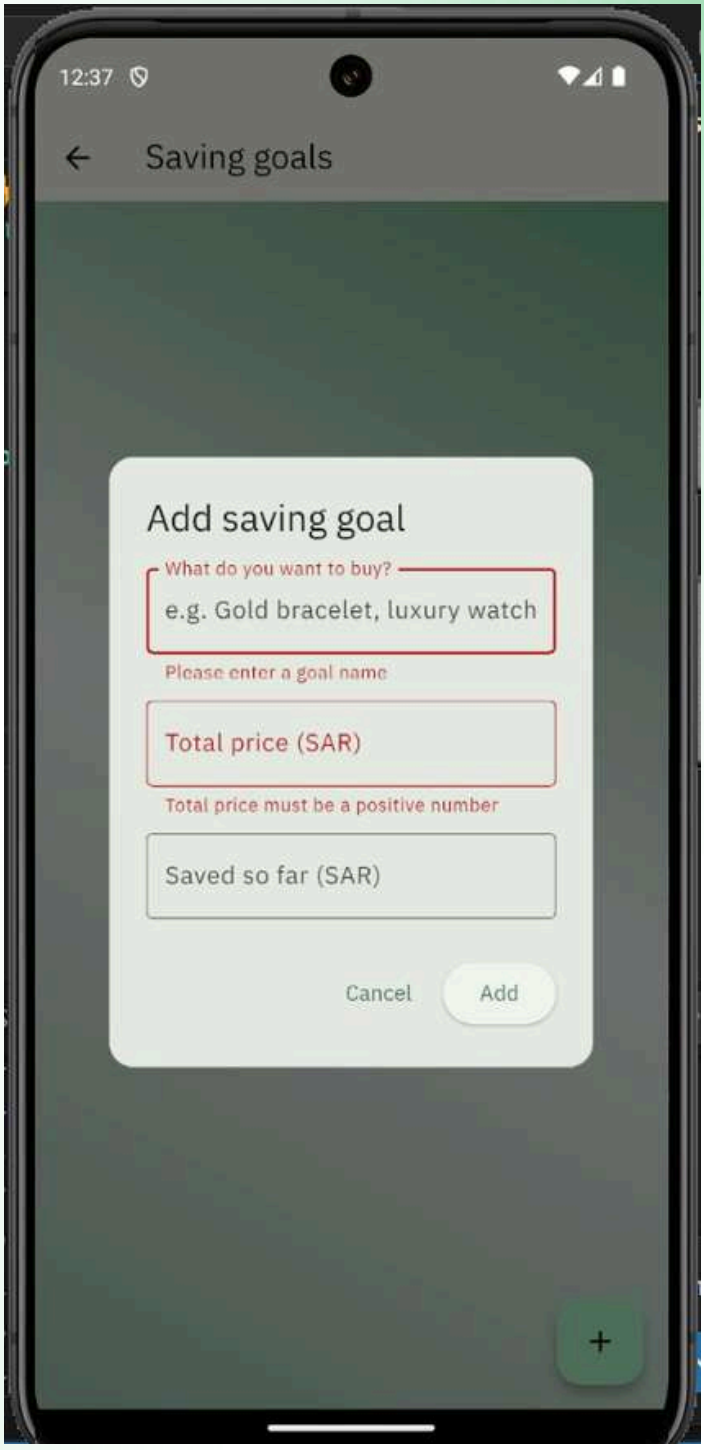
Scrum's Master comments based on the above questions:

Sprint Test Case

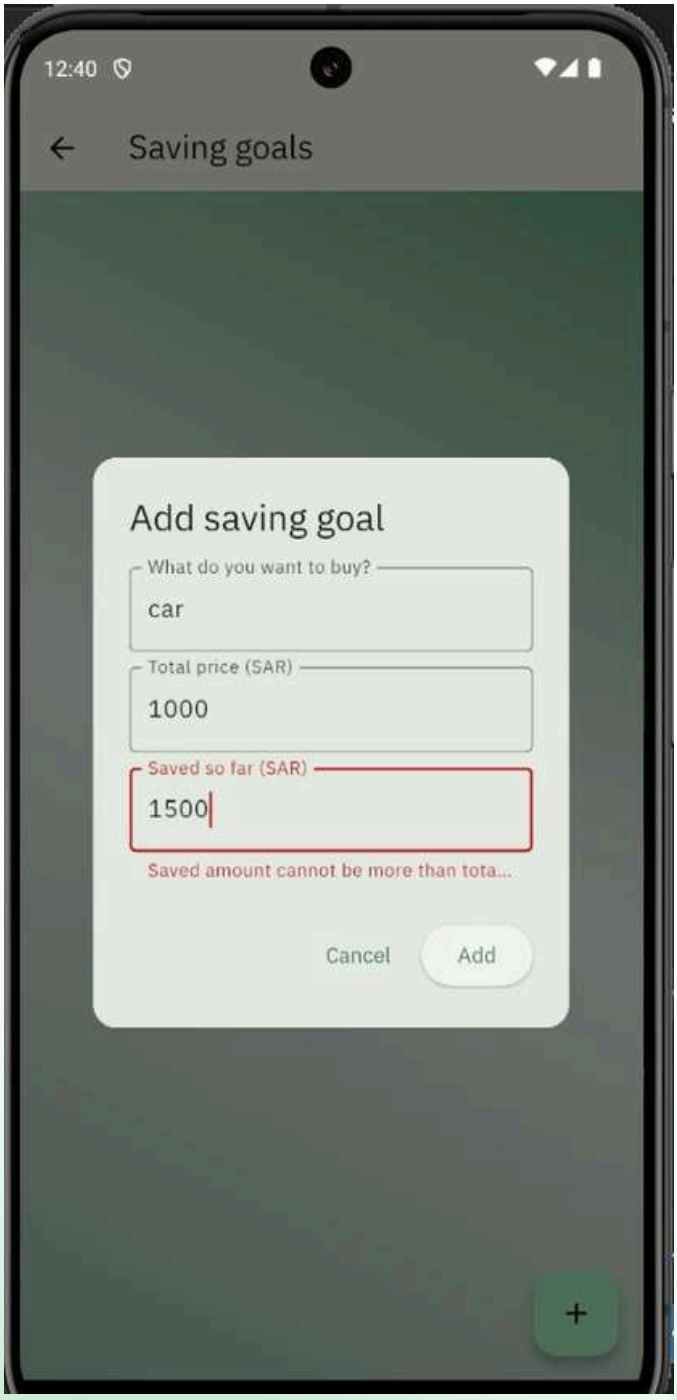
Test case No.	Test Case Description	Expected Results	Outcome (Pass / Fail / Comments)
TC01	User opens the Saving Goals screen while no goals exist	The empty state message appears: **"No saving goals yet. Tap + to add something you want to buy."**	pass
TC02	User taps **Add (+)**, goal name and total price empty, then taps **Add** in the dialog.	Red error text appears under required fields (Name + Total price). Goal is not added.	pass
TC03	User enters a valid total price (e.g., 1000) and a saved amount **greater** than total (e.g., 1500), then taps **Add**.	"Saved so far" field is highlighted in red with error message: **"Saved amount cannot be more than total price"**. Goal is **not** added.	pass
TC04	User enters a valid total price (e.g., 500) and a **negative** saved amount (e.g., -20), then taps **Add**.	"Saved so far" field is highlighted in red with error message: **"Saved amount cannot be negative"**. Goal is **not** added.	pass
TC05	User enters a valid name (e.g., "car"), total price 8000, and saved amount 2000, then taps **Add**.	Dialog closes. New goal card appears with: **Progress: 25%**, line "2000.00 / 8000.00 SAR", and "Remaining: 6000.00 SAR". Data is stored in saving_goals collection.	pass
TC06	For an existing goal "car" with total price 8000, user updates saved amount to 8000 using the edit icon.	Card shows **Progress: 100%**, "8000.00 / 8000.00 SAR", "Remaining: 0.00 SAR" and green message **"Goal completed! Great job 🎉"**.	pass

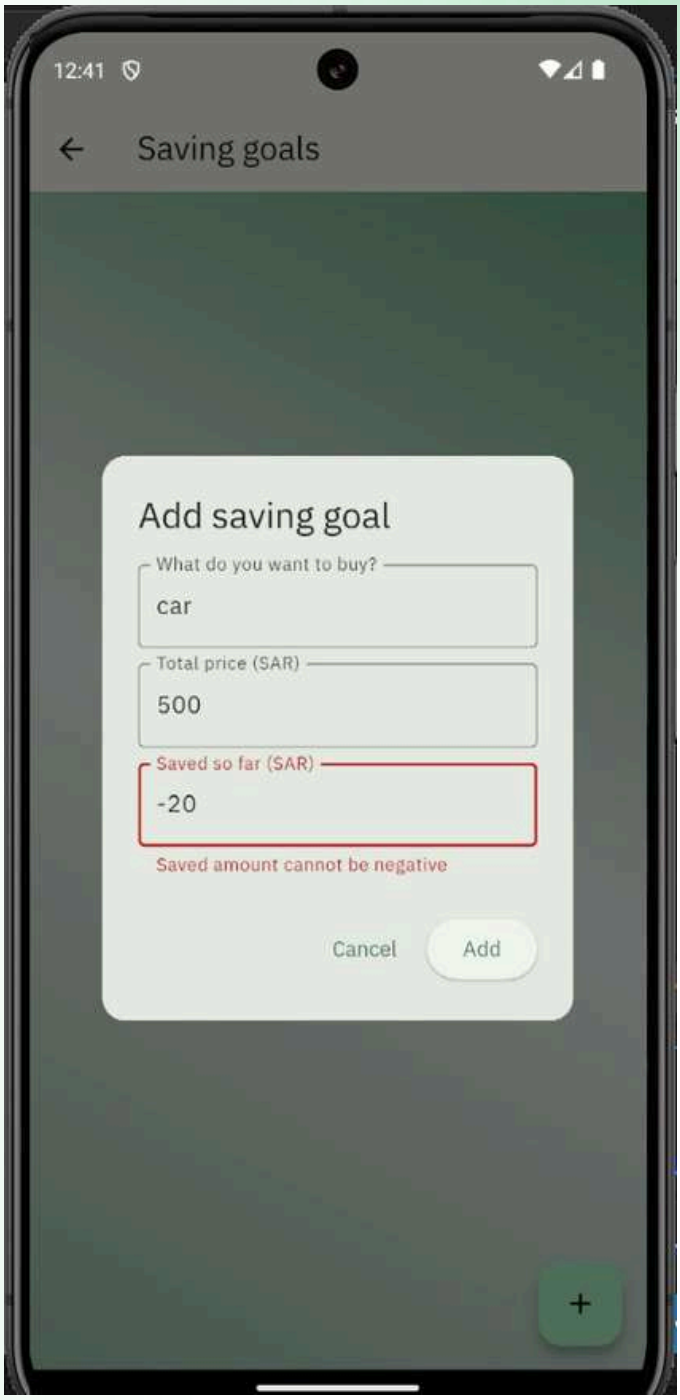
Sprint Test Case



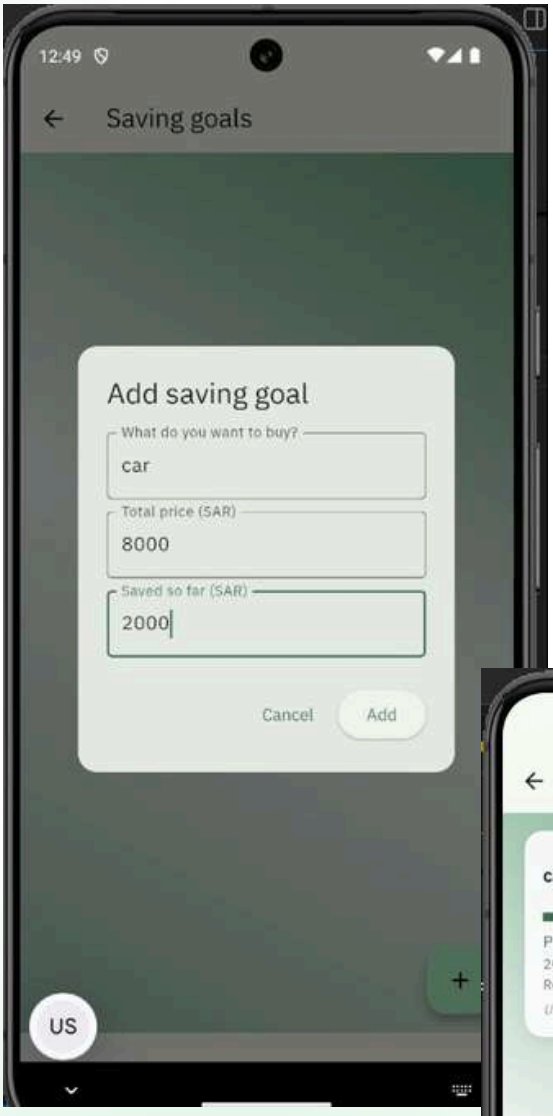


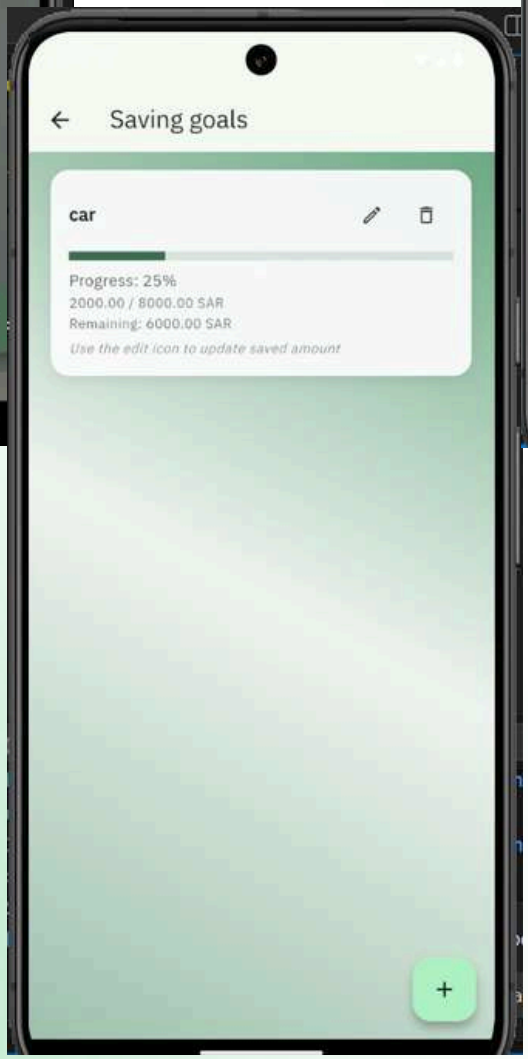
Sprint Test Case

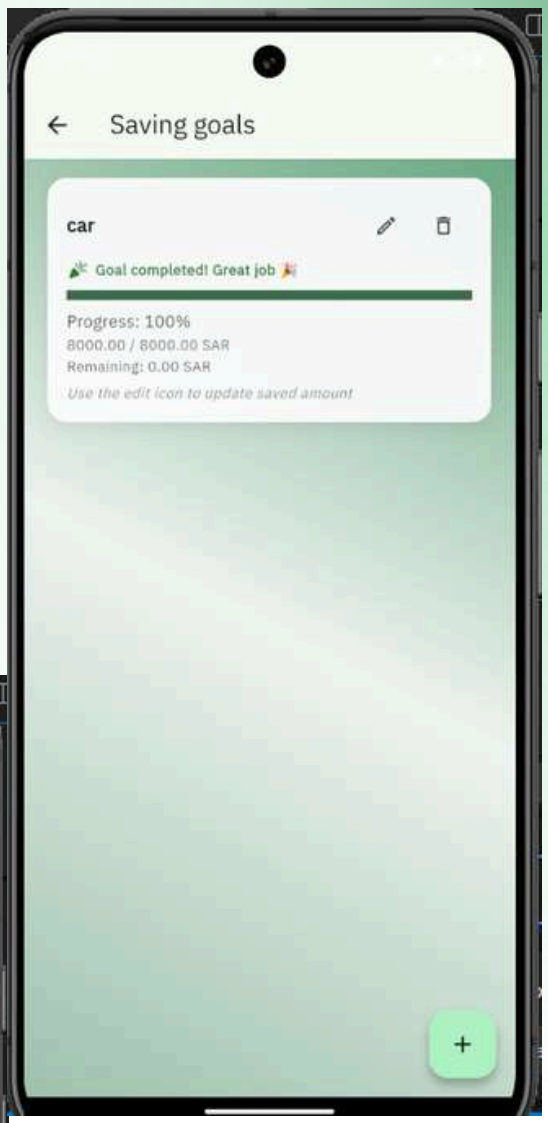




Sprint Test Case







Coding

EXPLORER

- MALI
  - m
    - .dart_tool
    - .idea
    - android
    - build
    - ios
    - lib
      - Screens
        - addBudget_screen.dart 1
        - home_screen.dart 1
        - login_screen.dart 1
        - my_daily_expenses_screen.dart
        - personal_information_screen.dart 2
        - saving_goals_screen.dart
        - sign_up_screen.dart
        - splash_screen.dart
      - main.dart 1
    - linux
    - macos
    - test
    - web
    - windows
- OUTLINE
- TIMELINE
- DEPENDENCIES

my_daily_expenses_screen.dart | login_screen.dart 1 | addBudget_screen.dart 1 | saving_g

m > lib > Screens > saving_goals_screen ... _showAddGoalDialog

```dart
import 'package:flutter/material.dart';
import 'package:google_fonts/google_fonts.dart';
import 'package:cloud_firestore/cloud_firestore.dart';

class SavingGoalsScreen extends StatefulWidget {
  const SavingGoalsScreen({super.key});

  @override
  State<SavingGoalsScreen> createState() => _SavingGoalsScreenState();
}

class _SavingGoalsScreenState extends State<SavingGoalsScreen> {
  final TextEditingController _nameController = TextEditingController();
  final TextEditingController _targetController = TextEditingController();
  final TextEditingController _savedController = TextEditingController();

  @override
  void dispose() {
    _nameController.dispose();
    _targetController.dispose();
    _savedController.dispose();
    super.dispose();
  }

  Future<void> _addGoal(
    String name,
    double targetAmount,
    double savedAmount,
  ) async {
```

EXPLORER

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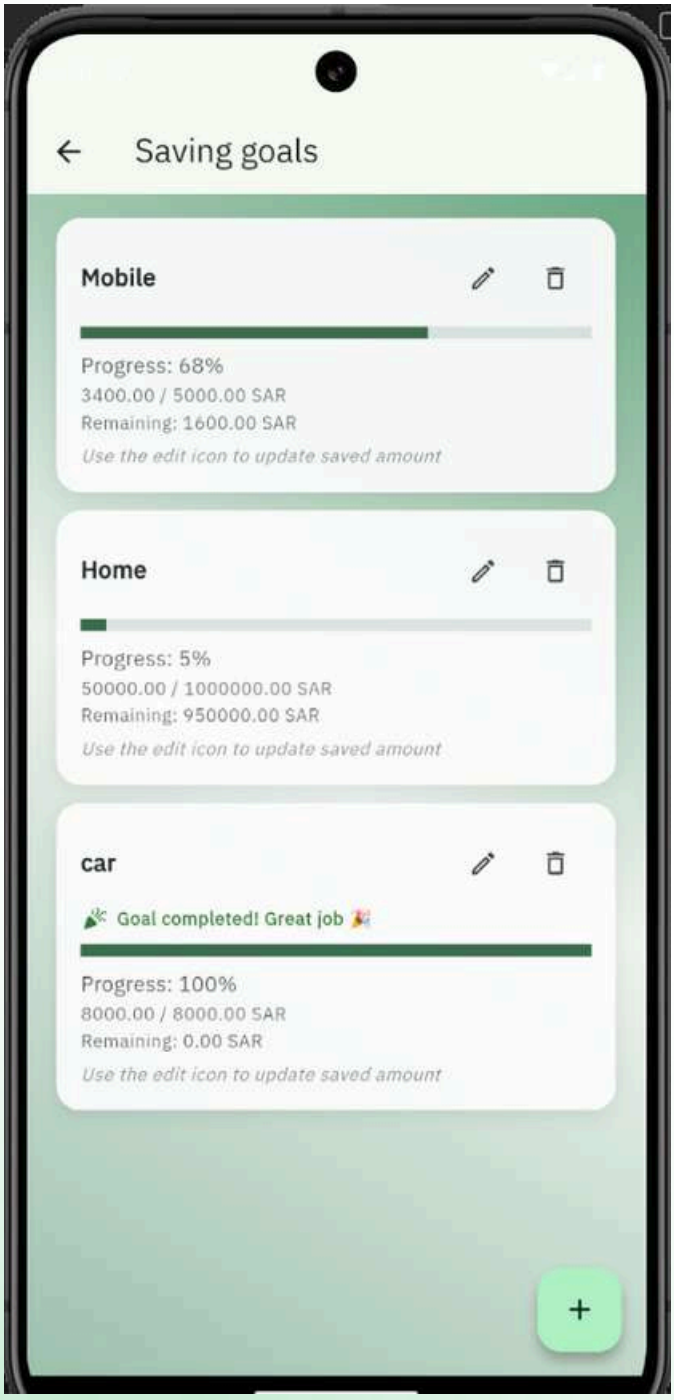
- OUTLINE
- TIMELINE
- DEPENDENCIES

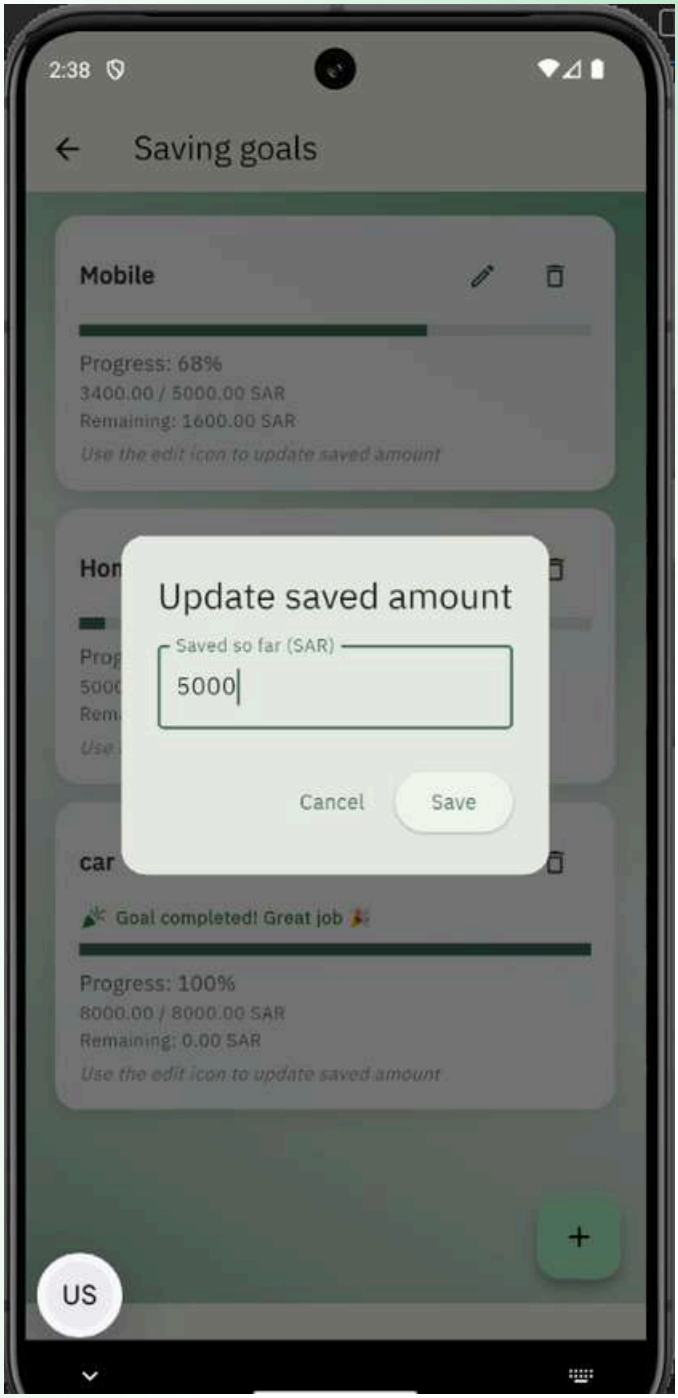
my_daily_expenses_screen.dart | login_screen.dart 1 | addBudget_screen.dart 1 | saving_goals_screen.dart

m > lib > Screens > saving_goals_screen ... _showAddGoalDialog

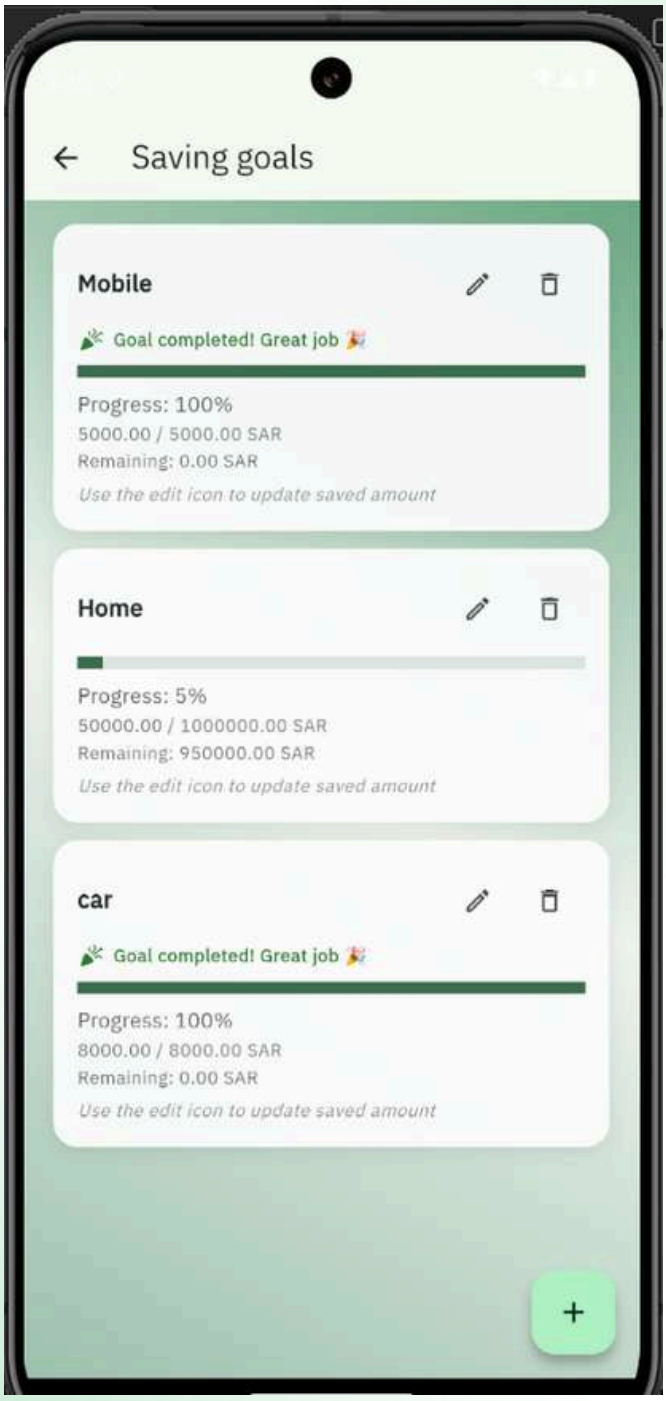
```dart
class _SavingGoalsScreenState extends State<SavingGoalsScreen> {
  Future<void> _addGoal(
    double savedAmount,
  ) async {
    await FirebaseFirestore.instance.collection('saving_goals').add({
      'name': name,
      'targetAmount': targetAmount,
      'currentAmount': savedAmount.clamp(0, targetAmount),
      'createdAt': Timestamp.now(),
    });

    _nameController.clear();
    _targetController.clear();
    _savedController.clear();
    if (mounted) Navigator.pop(context);
  }

  Future<void> _updateGoalSavedAmount(
    String docId,
    double targetAmount,
    double currentAmount,
  ) async {
    _savedController.text = currentAmount.toStringAsFixed(2);

    showDialog(
      context: context,
      builder: (context) {
        return AlertDialog(
          shape: RoundedRectangleBorder(
```

Interfaces


Saving goals
Mobile
Progress: 68%
3400.00 / 5000.00 SAR
Remaining: 1600.00 SAR
Use the edit icon to update saved amount
Home
Progress: 5%
50000.00 / 1000000.00 SAR
Remaining: 950000.00 SAR
Use the edit icon to update saved amount
car
Goal completed! Great job
Progress: 100%
8000.00 / 8000.00 SAR
Remaining: 0.00 SAR
Use the edit icon to update saved amount

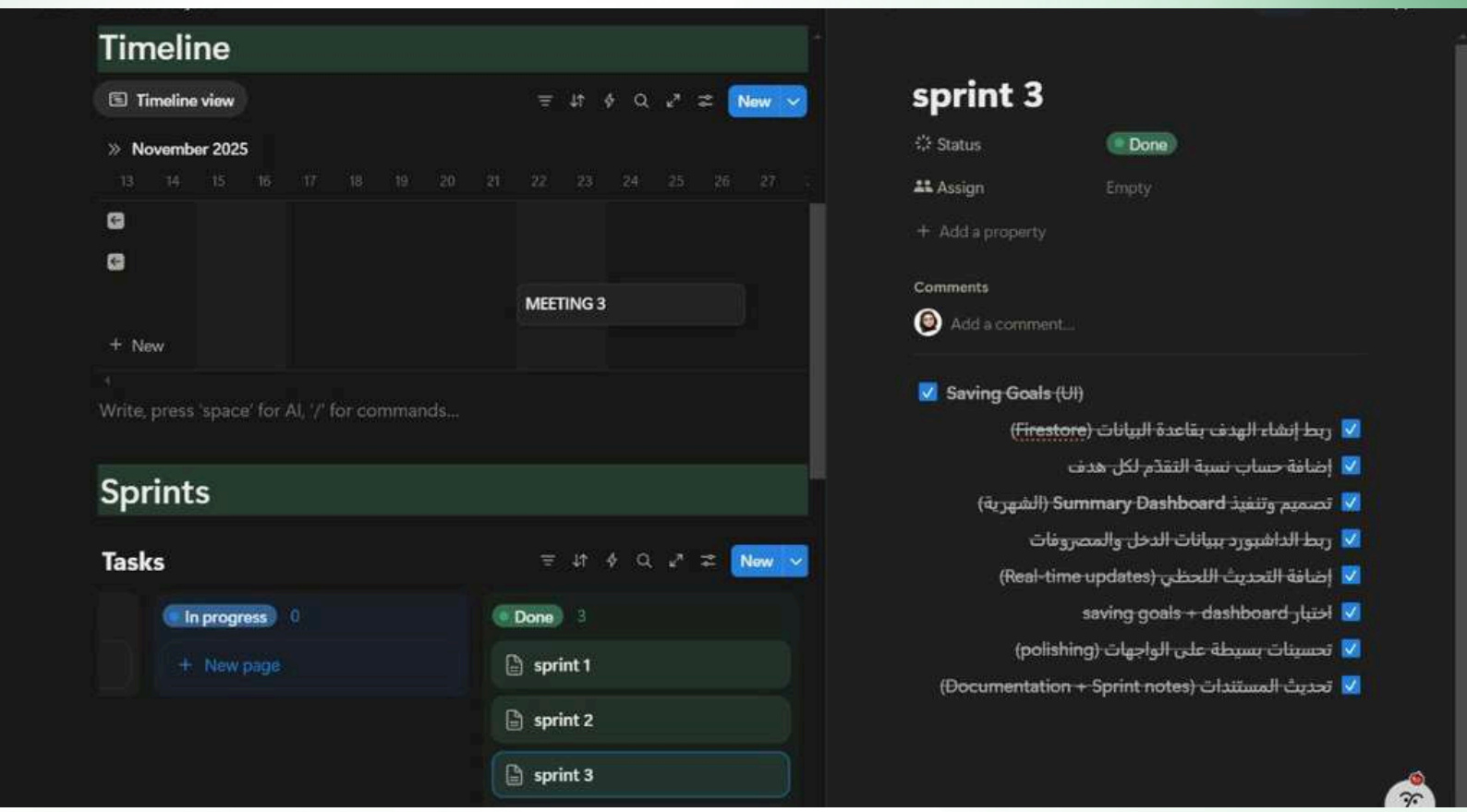

2:38
Saving goals
Mobile
Progress: 68%
3400.00 / 5000.00 SAR
Remaining: 1600.00 SAR
Use the edit icon to update saved amount
Update saved amount
Saved so far (SAR)
5000
Cancel
Save
car
Goal completed! Great job
Progress: 100%
8000.00 / 8000.00 SAR
Remaining: 0.00 SAR
Use the edit icon to update saved amount
US

Interfaces

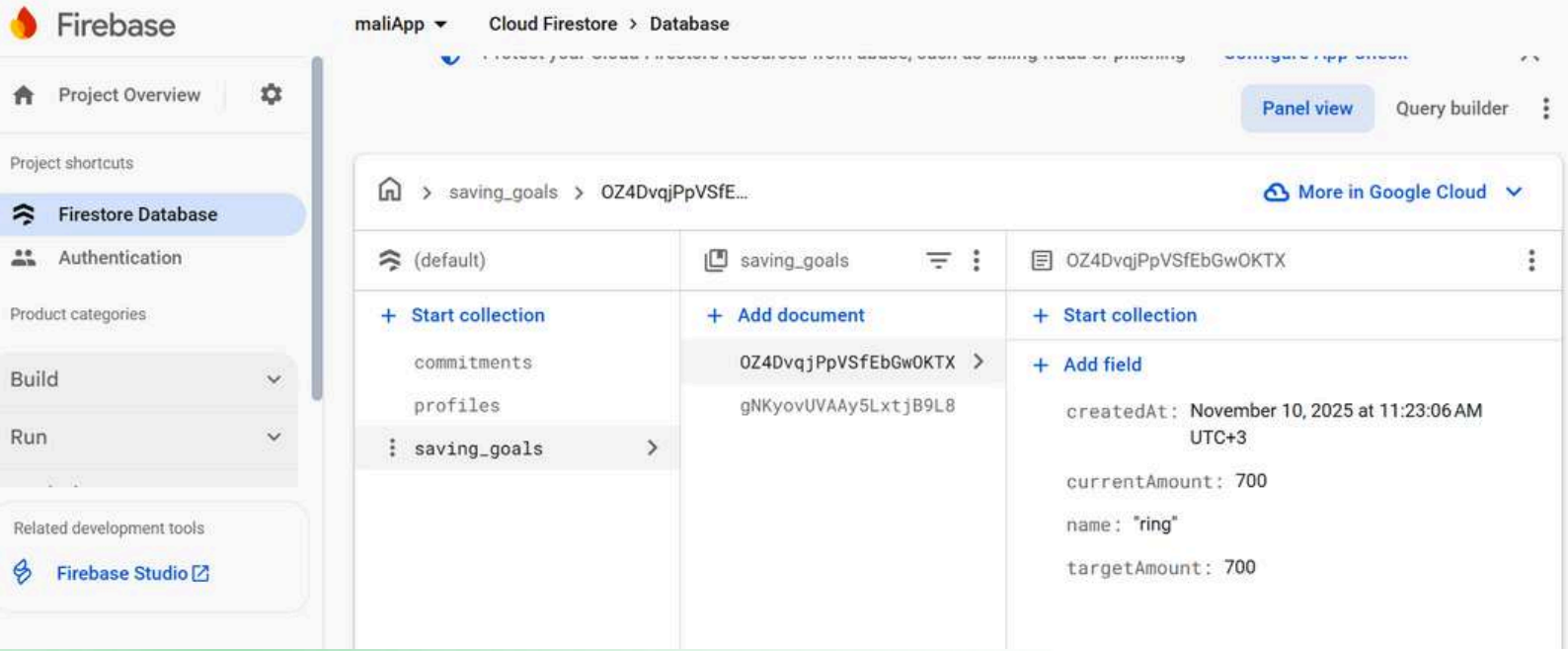

Saving goals
Mobile
Goal completed! Great job
Progress: 100%
5000.00 / 5000.00 SAR
Remaining: 0.00 SAR
Use the edit icon to update saved amount
Home
Progress: 5%
50000.00 / 1000000.00 SAR
Remaining: 950000.00 SAR
Use the edit icon to update saved amount
car
Goal completed! Great job
Progress: 100%
8000.00 / 8000.00 SAR
Remaining: 0.00 SAR
Use the edit icon to update saved amount
+

Tools

Notion



Firebase



Thank you